

Scaling Laws of Foundation Models: Paradigms, Theoretical Foundations, and Future Frontiers

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Foundation models have established empirical regularities linking computational resources, data volume, and predictive performance through power-law relationships. Recent transitions from dense parameter expansion to compute-optimal regimes represent a fundamental shift in how researchers allocate training budgets to maximize capability. However, approaching data scarcity limits, diminishing returns on brute-force scaling, and unpredictable emergent behaviors challenge the universality of existing formulations. This survey synthesizes five distinct scaling paradigms: parametric compute-optimality, architectural sparsity, data-centric composition, optimization dynamics, and downstream capability forecasting. Analysis reveals that future efficiency gains depend less on increasing parameter counts and more on decoupling active compute from total parameters while prioritizing data quality over quantity. Furthermore, the divergence between smooth pretraining loss curves and discontinuous downstream capability emergence indicates a critical need for task-specific scaling frameworks rather than relying solely on aggregate loss metrics. Sustainable progress in foundation model development now requires shifting focus toward test-time computation, synthetic data curricula, and mechanistic theoretical unification to navigate post-scaling constraints.

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1 Introduction

The historical trajectory of foundation model development has been defined by an exponential escalation in computational demand that fundamentally outpaces traditional hardware scaling laws [107, 162]. Contemporary research identifies a critical inflection point where brute-force parameter expansion encounters diminishing returns due to finite data availability and unsustainable economic costs. Consequently, the field is undergoing a structural transition from single-model optimization toward complex ecosystem architectures involving agentic swarms and physical interaction layers [107]. Establishing a unified theoretical framework for these shifting dynamics is essential to guide resource allocation beyond the era of unrestricted scale.

The core methodologies can generally be categorized into several primary paradigms. 1) Pre-training compute-optimal scaling establishes the fundamental relationship between resource allocation and

loss reduction, distinguishing dense expansion from efficiency-focused mechanisms [162, 186]. 2) Architectural sparsity approaches decouple active compute from total parameter count through mixture-of-experts dynamics, creating distinct scaling frontiers for sparse models [186]. 3) Alternative dimensions including quantization, compression, and retrieval-augmented generation serve as functional substitutes for raw parameter growth in post-training regimes [162, 186]. 4) Unified reference structures integrate these divergent paths to map specific scaling laws against economic constraints and deployment scenarios [162].

Despite the predictive success of aggregate loss reduction, a fundamental contradiction persists where these smooth trajectories fail to forecast specific capability transitions or safety risks [24]. Current development paths reveal an adversarial dynamic in which exponential cost growth collides with the impending exhaustion of high-quality training data sources [162]. This tension has ignited a technological arms race pitting brute-force parameter expansion against

alternative efficiency paradigms designed to bypass physical limits. Furthermore, the inability to guarantee appropriate model behavior despite superhuman performance metrics highlights a critical divergence between raw capability scaling and effective control mechanisms [24].

Fragmented empirical findings across isolated dimensions such as architecture, data composition, and inference strategies currently prevent the formation of a cohesive framework for guiding real-world deployment under strict resource constraints [186]. Traditional power-law formulations frequently fail to capture the complex dynamics of sparse models, mixture-of-experts systems, and retrieval-augmented architectures where active compute decouples from parameter count, leading to broken scaling predictions in modern regimes. The transition toward agentic ecosystems and physical AI introduces hierarchical layers that current theories do not adequately address, particularly regarding energy efficiency and context memory dynamics [107]. A critical synthesis is required to bridge the gap between theoretical scaling predictions and the practical limitations imposed by hardware bottlenecks and undisclosed hyperparameters in proprietary systems. Therefore, the primary motivation of this survey is to synthesize these fragmented observations into a unified taxonomy that guides resource allocation, predicts capability limits, and identifies sustainable pathways for future model development beyond brute-force parameter expansion.

Contributions. In this survey, we make three key contributions: **1) Unified Macro-Scope Taxonomy:** We synthesize fragmented scaling observations across parametric, architectural, data-centric, optimization, and capability paradigms into a cohesive framework that guides resource allocation and predicts capability limits as detailed in Sec. 3. **2) Precise Boundary Definition:** The work rigorously distinguishes between pre-training compute-optimal regimes, architectural efficiency mechanisms, and downstream forecasting frameworks to clarify the specific domain of applicability for each scaling law variant discussed in Sec. 2 and Sec. 3. **3) Strategic Development Roadmap:** We provide a structured path for future model development that transitions focus from brute-force parameter expansion toward algorithmic innovation and theoretical unification, addressing the sustainability and predictability challenges outlined in Sec. 4.

Scope. Centered on the synthesis of fragmented observations into a unified taxonomy, this survey delineates its coverage across five core scaling paradigms: parametric compute-optimal relationships as detailed

in Sec. 3.1, architectural efficiency via sparsity mechanisms explored in Sec. 3.2, data-centric composition strategies analyzed in Sec. 3.3, training dynamics optimization discussed in Sec. 3.4, and downstream capability forecasting examined in Sec. 3.5. We strictly exclude granular engineering implementation details, hardware-specific kernel optimizations, and non-scaling-related application case studies to maintain focus on theoretical and empirical scaling laws rather than cataloging individual model releases or benchmark leaderboards. This boundary enforcement ensures the analysis guides resource allocation through principled understanding instead of documenting transient system specifications or isolated performance metrics.

Survey Pipeline. Following the foundational context established in Sec. 2, which details the mathematical formulations and historical evolution of scaling hypotheses, the narrative advances to a systematic deconstruction of methodological paradigms in Sec. 3. This core segment categorizes research into five distinct domains, ranging from parametric compute-optimal relationships and architectural efficiency strategies to data-centric composition, optimization dynamics, and downstream capability forecasting. The discussion concludes in Sec. 4 by synthesizing macro-dimensional limitations concerning sustainability and predictability before mapping concrete future directions for algorithmic innovation and governance frameworks. Finally, Sec. 5 offers a concise synthesis of the unified theory and remaining gaps identified throughout the review.

2 Background

This chapter establishes the foundational context required to interpret neural scaling law literature, beginning with the core mathematical formulations that define power-law relationships between loss, model size, and compute budgets in Sec. 2.1. We then trace the historical evolution of these concepts from early empirical observations to modern compute-optimal frameworks in Sec. 2.2, followed by an examination of the evaluation protocols and benchmarking standards used to validate scaling hypotheses in Sec. 2.3. The discussion extends to adjacent research domains, including optimization theory and information theory, which provide theoretical grounding for observed scaling behaviors as detailed in Sec. 2.4. Finally, we synthesize the current status of background knowledge to identify existing gaps that motivate further investigation.

2.1 Problem Definition

Asymptotic Loss Decomposition. The fundamental behavior of neural scaling is governed by a power-law decay in validation loss as model parameters N and training tokens D increase, asymptotically approaching a non-zero constant E that reflects the inherent entropy of the data distribution [72, 205]. This relationship decomposes the total loss into an irreducible pedestal and reducible components driven by parameter and data efficiency, formally expressed as

$$L(N, D) = E + \frac{A}{N^\alpha} + \frac{B}{D^\beta} \quad (1)$$

where exponents α and β quantify the marginal returns of scaling. In the lazy training regime, theoretical analysis links these decay rates to the spectral properties of the Neural Tangent Kernel. Such work derives the temporal loss exponent ξ from the eigenvalue spectrum λ_n , target function regularity, and input dimensionality [218]:

$$L(t) \sim t^{-\xi}, \quad \xi = \frac{\kappa}{\nu} \quad (2)$$

Recent investigations further reveal that the dominance of the irreducible term E suppresses observed scaling exponents in large language models. These studies establish a lower bound on efficiency determined by the intrinsic dimension d of the data manifold and its Hurst exponent h [205]:

$$\alpha \geq \frac{4h}{d} \quad (3)$$

These mathematical constraints delineate the boundary between reducible optimization error and the intrinsic limits imposed by data structure. They provide the theoretical substrate for the compute-optimal regimes discussed in subsequent sections.

Theoretical Determinants of Scaling Exponents.

Geometric constraints on the input space fundamentally dictate the rate at which generalization error decays, with theoretical derivations positing that the scaling exponent α is inversely proportional to the intrinsic dimension d of the data manifold [189]. Under assumptions of Lipschitz continuity and piecewise linear approximation, this relationship manifests as

$$L(N) \sim N^{-\alpha}, \quad \text{where } \alpha \approx \frac{4}{d}. \quad (4)$$

Complementary statistical mechanics frameworks extend this analysis to the spectral domain, demonstrating that power-law eigenvalue spectra $\lambda_k \sim k^{-\beta}$ in data covariance matrices induce corresponding convergence rates in student-teacher setups [229]. In such regimes, the generalization error ϵ_g scales with

the number of samples α according to the spectral decay rate

$$\epsilon_g \propto \alpha^{-\frac{\beta}{1+\beta}}, \quad (5)$$

indicating that the bottleneck for learning shifts from model width to the distinct eigenvalues L governing the order parameter dynamics. Rigorous examination of random feature models further confirms that non-linear monomial activations of degree p preserve these spectral slopes subject only to polylogarithmic corrections, ensuring the exponent remains stable across architectural variations [160]. Finally, percolation theory unifies these continuous and discrete perspectives by modeling data clusters near criticality, where the fractal dimension D of connected components determines the rank-size distribution and distinguishes between subcritical and supercritical learning regimes [25].

Unified Optimization Frontiers. Reconciling conflicting parameter-centric and data-centric regimes requires framing loss minimization as a constrained optimization problem governed by a Max-Bottleneck principle across capacity, coverage, and optimization constraints [86, 257]. This formulation defines an effective frontier k_* within the pattern rank space, where the total loss decomposes into irreducible components, misspecification error driven by the Zipfian tail index α , and estimation error modulated by optimization bias β . Under a fixed compute budget $C = n \cdot t$, the optimal allocation of parameters and data steps is approximated via the Lambert W function, balancing these competing error sources. Theoretical analysis of stochastic gradient dynamics further reveals distinct phase transitions in this compute-optimal frontier, where the dominant constraint shifts between model capacity, feature distortion, and optimizer noise spectral components [159]. Derived from Volterra equations using Power-Law Random Features (PLRF) and deterministic equivalents, these transitions distinguish between point mass and absolutely continuous bulk phases, identifying a universal scaling exponent of 0.5 and suggesting an optimal batch size order of $O(1)$ when forcing functions dominate kernel interactions. Extending beyond monotonic power laws, information-theoretic formulations map model-data interactions to channel bandwidth and signal power to capture non-monotonic degradation regimes [236]. By applying the Shannon-Hartley theorem, the loss is expressed as the reciprocal of channel capacity $L = 1/C_{LLM}$, where bandwidth scales as N^α and signal power as D^β , subject to additive joint noise terms $c(DN)^\gamma$ and data-induced noise dD^δ . This approach successfully models U-shaped performance curves and achieves high fidelity with a pooled R^2 metric, demonstrating how resource

magnitude jointly influences signal-to-noise ratios in large-scale training.

2.2 Technological History and Roadmap

Eras of Compute Accumulation. Tracing the trajectory of computational expenditure reveals distinct regimes governing machine learning development prior to the formulation of explicit scaling laws. During the pre-deep learning period from 1952 to 2010, system complexity expanded in alignment with hardware capabilities, exhibiting a compute doubling time of 17 to 29 months consistent with Moore’s Law [188]. A sharp discontinuity appeared around 2010 with the onset of deep learning, where the doubling period accelerated to between 4 and 9 months, significantly outpacing underlying hardware improvement rates. This acceleration evolved further circa 2015 into a separate Large-Scale Era, characterized by a baseline compute shift of two to three orders of magnitude and a distinct growth rate with a 9 to 10 month doubling time that operates independently of earlier trends.

Quantification of Efficiency Gaps. Empirical investigations reveal that field-level progress in scaling models incurs computational costs significantly exceeding theoretical lower bounds or ablation-based estimates, indicating a persistent efficiency gap [213]. Quantitative regression across diverse domains demonstrates super-linear polynomial scaling exponents where computation grows drastically faster than performance improvements, with log-log analyses yielding exponents of 12.5 for ImageNet and 37.2 for Named Entity Recognition tasks. This divergence between algorithmic efficiency gains, historically improving at a rate of roughly ten times every three years, and the inflation of total hardware burden suggests that current deep learning trajectories face unsustainable economic and environmental constraints driven by cost structures resembling fourth-order polynomials.

Compute-Optimal Regimes and Hyperparameter Stability. Moving beyond fixed-dataset protocols, recent work establishes that optimal performance requires aligning token counts with parameter scales to maximize compute efficiency [47]. This shift identifies a target ratio of approximately 20 tokens per parameter, correcting previous tendencies to under-train large models on static corpora. Achieving such scaling demands stable hyperparameter transfer across orders of magnitude, a challenge addressed by maximal update parameterization (μP) which reduces scaling variance compared to standard initialization

schemes. Empirical validation on wafer-scale clusters confirms that dense architectures can adhere to these theoretical efficiency bounds without architectural modification, demonstrating linear scaling behavior and improved zero-shot accuracy when trained under these compute-optimal conditions.

2.3 Datasets, Metrics, and Benchmarks

From Static Snapshots to Dynamic Trajectories.

Early assessment frameworks often relied on isolated accuracy metrics captured at fixed scales, a practice that obscured the distinction between genuine architectural advantages and mere disparities in compute allocation until researchers adopted integral fitness functions combining quality, throughput, and memory costs [134]. Contemporary methodologies have since shifted toward deriving full power-law scaling curves across dense compute grids, enabling robust comparisons that remain invariant to specific training checkpoints by fitting functions of the form $\mathcal{L}(C) = A_c \cdot (C + B_c)^{-\alpha_c} + E_c$ [152]. Recent advances further refine these trajectories by modeling post-training capability boundaries as monotone sigmoid functions of log-compute, thereby distinguishing static knowledge limits from evolving reasoning frontiers through techniques such as smoothed pinball loss minimization and high conditional quantile estimation [245].

Dataset Composition and Quality Impact. Empirical analyses demonstrate that high-quality, domain-aligned corpora consistently outperform larger, noisier, or multilingual mixtures across model scales, challenging the assumption that raw token volume alone drives scaling efficiency [151, 225]. In specialized domains such as remote sensing and biomedicine, performance gains correlate strongly with geographic diversity, modality heterogeneity, and task complexity rather than backbone capacity increases, often exhibiting flat parameter scaling slopes where $b \approx 0$ [114, 228]. Dynamic data management strategies during fine-tuning further reveal that instruction quality and complexity yield superior scaling returns compared to static dataset expansion, necessitating adaptive pruning loops over fixed corpus accumulation [225]. Moreover, dataset rankings remain stable across varying compute budgets and model sizes, suggesting that early-stage curation decisions fundamentally constrain the upper bounds of achievable performance regardless of subsequent architectural scaling [114, 151].

Alternative Efficiency Metrics. Beyond standard loss curves, Information Capacity provides a refined efficiency measure by integrating tokenizer compres-

sion gains against logarithmic inference costs to expose architectural disparities invisible to raw perplexity [243]. This approach normalizes *FLOPs* and utilizes arithmetic coding principles to achieve a Pearson correlation exceeding 0.98, effectively decoupling Mixture-of-Experts activation patterns from simple parameter counts. For parameter-efficient fine-tuning, the Mutual Information Upper Bound substitutes unstable cross-entropy measurements with Jensen-Shannon divergence estimates, tracking dependency dynamics between frozen priors and adapted knowledge through hidden state analysis [247]. Such metrics exhibit a monotonic decrease trend that correlates inversely with generalization capability, offering internal stability where traditional power-law fits frequently diverge across varying adaptation ranks. By employing negative offset normalization and single-reference prediction, these bounds mitigate Cross-Entropy spikes and reduce NLL prediction errors to below 3%, delivering scale-invariant signals consistent even under black-box API limitations [243, 247].

Domain-Specific Benchmarking Constraints. Extrapolating performance plateaus in robotics and medical imaging requires distinct methodologies compared to internet-scale pretraining, primarily due to severe data scarcity that limits the density of empirical observations [150, 178]. In these fields, evaluation often hinges on zero-to-nonzero transitions in robotic task success rates across seen and unseen distributions, or on identifying specific ROC-AUC thresholds in chest X-ray analysis via linear probe fine-tuning on frozen encoder architectures. Such approaches necessitate estimating critical sample sizes n_o where performance gains saturate, rather than relying on continuous power-law curves derived from massive corpora. Parallel challenges appear in black-box optimization, where the construction of large-scale trajectory datasets containing over 557K samples enables the verification of power-law relationships through decoder-only transformer pre-training [93]. Although trajectory imitation fidelity and constrained decoding allow researchers to fit error rate models of the form $ErrorRate(X) = (K/X)^\theta + E$, the resulting compute scaling exponent γ often remains shallower than those observed in language modeling. This discrepancy highlights a persistent gap between synthetic benchmark coverage and the rigorous generalization requirements needed for real-world AutoML applications.

2.4 Related Research Domains

Spectral Foundations and Kernel Dynamics. Grounded in the Neural Tangent Kernel framework,

the asymptotic convergence rate of loss functions is dictated by the eigenvalue distribution of the limiting kernel operator, where a decay profile $\lambda_n \sim n^{-\nu}$ determines the universal exponent ξ via Birman-Solomjak theory [218]. This spectral perspective reveals a bifurcation into variance-limited and resolution-limited regimes, with the latter governed by the intrinsic dimension of the data manifold and the duality between dataset size and parameter count [10]. Moving beyond lazy training approximations, statistical mechanics approaches utilizing student-teacher frameworks demonstrate how power-law data spectra induce specific generalization error scaling through the dynamics of macroscopic order parameters, marking a transition from exponential to power-law convergence [229]. Recent extensions into the feature learning regime further employ state evolution and dynamical mean field theory to characterize excess risk and weight spectrum tails, distinguishing between benign and harmful overfitting through the analysis of spectral bulk versus spikes in deep linear and diagonal networks [21, 44].

Information-Theoretic Bounds and Compression Equivalence. Fundamental limits on predictive performance arise from the equivalence between optimal prediction and lossless compression, where the ideal model size balances parameter overhead against data redundancy [45]. This perspective allows information-theoretic error decomposition to separate reducible loss into estimation and misspecification components, deriving linear compute-optimal scaling relations from first principles under fixed FLOP budgets [86, 87]. Beyond aggregate bounds, concept learning frameworks map skill acquisition to LDPC decoding thresholds on bipartite graphs, explaining observed phase transitions and performance plateaus through the connectivity properties of random graphs [149]. Complementing these discrete models, conditional entropy reduction links auto-regressive information gain to ridge regression uncertainty, offering a geometric interpretation of token saliency and scaling behavior via posterior variance analysis [209].

Statistical Mechanics and Phase Transition Analogies. Deep network initialization dynamics exhibit universality classes of non-equilibrium statistical mechanics, where signal propagation maps to absorbing phase transitions and crackling noise phenomena characterized by Barkhausen noise exponents τ_s, τ_d and susceptibility peaks χ within the directed percolation class $(1+1)D$ [60, 208]. The geometry of data distributions further governs scaling regimes through percolation criticality, unifying discrete subtask learning and manifold approximation via cluster fractal dimensions D and rank-size distributions that deter-

mine the loss exponent α [25]. Renormalization group flows adapted for discrete spectra explain the convergence of non-Gaussian corrections toward Gaussian Process fixed points as dataset size increases, utilizing Wilsonian coarse-graining to identify UV irrelevance and mass-like ridge parameters near the learnability threshold [38]. Universal diversity scaling laws arise from linear innovation-dependence mechanisms or latent heterogeneity, linking fluctuation statistics to growth dynamics in generative systems through Heaps’ law and quadratic fluctuation scaling derived from the law of total variance [140]. Semantic landscape models formalize training trajectories as thermally activated hops across barrier distributions, deriving grokking and emergence as percolation transitions on heuristic graphs governed by Arrhenius rates and scale-dependent connectivity thresholds [61].

Optimization Dynamics and Implicit Bias. Training trajectories evolve through distinct dynamical regimes where spectral complexity norms govern generalization error, initially following dataset-independent power laws before saturating according to data volume [42]. Dynamical mean field theory exposes an asymmetry between time and model size scaling exponents, revealing that finite-width corrections and overfitting gaps accumulate gradually via repeated data reuse integrals rather than appearing instantaneously [22]. Under gradient descent, parameter counts and training iterations function as interchangeable resources whose product dictates the onset of generalization, thereby explaining non-monotonic double descent phenomena through varying rates of noisy feature acquisition [20]. Furthermore, finite networks exhibit power-law aging in training response magnitude instead of reaching static equilibrium, a process that drives input space splitting where majority classes dominate early learning phases prior to the separation of minority clusters [23, 147].

2.5 Discussion

Despite the proliferation of empirical power-law fits, a fundamental disconnect persists between these curve-fitting exercises and mechanistic theoretical derivations, leaving the field without a unified framework capable of predicting scaling behaviors across diverse architectural regimes. This theoretical fragmentation directly compromises extrapolation reliability, as fitted exponents exhibit high sensitivity to variations in data quality and optimization hyperparameters, resulting in confidence intervals that often fail to capture true performance trajectories. Furthermore,

standard compute-optimal training assumptions frequently overlook the non-monotonic impact of dataset contamination on convergence, where noise introduces irregularities in the irreducible loss floor that simple parametric models cannot anticipate. Consequently, the reliance on smooth functional forms for $L(N, D)$ masks underlying optimization gaps, suggesting that current scaling laws may serve more as descriptive heuristics than predictive physical laws until mechanistic drivers are rigorously integrated.

3 Methods

This section categorizes scaling law research into five distinct conceptual paradigms to analyze algorithmic mechanisms and bottlenecks. [Sec. 3.1](#) establishes the fundamental relationships between model parameters, data tokens, and compute budgets under dense assumptions. [Sec. 3.2](#) examines how architectural modifications such as sparsity and mixture-of-experts decouple parameter counts from active compute requirements. [Sec. 3.3](#) shifts focus to data properties, investigating how quality and composition influence scaling trajectories beyond simple token metrics. [Sec. 3.4](#) analyzes optimization hyperparameters and loss dynamics during training scale increases. Finally, [Sec. 3.5](#) extends these principles to downstream tasks and emergent capabilities like reasoning and code generation.

3.1 Parametric and Compute-Optimal Scaling

This section establishes the foundational principles governing the interplay between model parameters, training data volume, and computational cost under dense architecture assumptions. We begin by defining the standard power-law formulations that characterize loss reduction as a function of scale in [Sec. 3.1.1](#), followed by an examination of IsoFLOP analysis methods used to map trade-offs within fixed compute budgets in [Sec. 3.1.2](#). The discussion then progresses to the derivation of compute-optimal frontiers that identify ideal scaling proportions in [Sec. 3.1.3](#), alongside techniques for extrapolating small-scale results to larger regimes in [Sec. 3.1.4](#). Finally, we address critical methodological nuances, including variations in parameter counting conventions in [Sec. 3.1.5](#) and the existence of broken power laws that signal phase transitions in scaling behavior in [Sec. 3.1.6](#).

Table 1 Taxonomy of representative methods in **Parametric and Compute-Optimal Scaling**. This table presents a taxonomy of scaling methodologies, shows derivation modes and optimization targets, and highlights the shift from empirical fitting to analytical mechanisms.

Method	Venue	Derivation Mode	Optimization Target	Highlight
Power-Law Formulations (Sec. 3.1.1)				
M4 Estimator [4]	NeurIPS’22	Convex Combination	Asymptotic Error	M4 formulation blends random-guessing loss with asymptotic power laws to model non-asymptotic learning regimes.
Scaling Law Analysis [239]	ICLR’24	Domain-Specific Fitting	OOD Generalization	Validates consistent power-law scaling for time-series forecasting across in-distribution and out-of-distribution evaluation settings.
Code Scaling Law Fitting [127]	ACL’25	Empirical Code Scaling	Code-NL Mixture	Derives code-specific scaling laws revealing exponentially growing optimal data-to-parameter ratios distinct from natural language.
Tunable Complexity Seq [11]	arXiv’26	Empirical Regression	Graph Complexity	Demonstrates power-law emergence in synthetic graph data independent of input statistics, utilizing neural network regression fits.
Neural Scaling Hypothesis [119]	arXiv’26	Theoretical Decomposition	Universality Class	Posits universal fixed exponents driven by architectural mechanisms while attributing data specificity solely to coefficient variations.
IsoFLOP Analysis (Sec. 3.1.2)				
Compute-Optimal Scaling [76]	NeurIPS’22	Empirical Power Law	Compute Efficiency	Derives equal scaling exponents for parameters and tokens under fixed compute budgets.
Corrected IsoFLOP Scaling [167]	NeurIPS’24	Corrected IsoFLOP	Hyperparameter Sweep	Corrects IsoFLOP fitting biases via last-layer FLOP inclusion and scaled warmup durations.
IsoFLOP Surface Analysis [1]	ICML’25	3D Polynomial Surface	Sparsity Trade-off	Models 3D IsoFLOP surfaces optimizing total parameters and sparsity under fixed compute.
CO [172]	ACL’25	IsoFLOP Polynomial	Skill Specificity	Quantifies validation set bias shifting IsoFLOP optima across skill-specific scaling regimes.
Time-Constrained Scaling [118]	arXiv’26	Time-Budget Empirical	Wall-Clock Time	Derives steeper time-constrained scaling exponents distinct from standard compute-optimal IsoFLOP curves.
Compute-Optimal Frontiers (Sec. 3.1.3)				
Compute-Optimal Scaling [76]	NeurIPS’22	Empirical Power Law	Compute Efficiency	Derives equal scaling exponents for parameters and tokens to minimize loss under fixed FLOP budgets.
Inference Scaling Laws [177]	ICML’23	Analytical Optimization	Inference Cost	Optimizes model size and training tokens by minimizing total FLOPs conditioned on lifetime inference demand.
WSD [80]	arXiv’24	Wind Tunnel Experiment	Data-Model Ratio	Identifies a 192:1 data-to-model ratio using Warmup-Stable-Decay schedulers for continuous compute-efficient training.
Extrapolation Techniques (Sec. 3.1.4)				
M4 Estimator [4]	NeurIPS’22	Convex Combination	Asymptotic Error	M4 combines random-guessing loss and asymptotic power laws to minimize extrapolation error across non-asymptotic regimes.
AtS [109]	ICML’24	Phase Transition Detection	Fine-tuning Data	AtS integrates pre-learned data size to detect phase transitions, enabling accurate fine-tuning extrapolation from sparse curves.
Cross-Family Scaling Law [35]	ICML’24	Meta-Analytic Fitting	Cross-Family Loss	Fitting Hoffmann-style functions to intermediate checkpoints reduces noise and improves large-scale performance prediction accuracy.
Parameter Counting Conventions (Sec. 3.1.5)				
Pearce et al. [163]	TMLR’24	Analytical Simulation	Parameter Counting	Derives analytical relation showing embedding exclusion biases small-scale scaling exponents.
Corrected IsoFLOP Scaling [167]	NeurIPS’24	Corrected IsoFLOP	Hyperparameter Sweep	Corrects parameter definitions and warmup schedules to align Kaplan exponents with Chinchilla.
Scaling Law Checklist [104]	arXiv’25	Reproducibility Survey	Procedural Variance	Quantifies how embedding counting variations yield divergent optimal data-to-parameter ratios.
Broken Power Laws (Sec. 3.1.6)				
BNSL [27]	ICLR’22	Smoothly Broken	Double Descent	Defines smoothly broken power laws with tunable sharpness to model non-monotonic double descent phenomena.
Multilinear Skill-Basis [148]	NeurIPS’24	Analytical Multilinear	Skill Dynamics	Derives exact analytical expressions for sigmoidal skill emergence using orthogonal basis functions with power-law frequencies.
UNSL [28]	arXiv’26	Multivariate Broken	Performance Limits	Formulates multivariate broken laws integrating oppositional forces and smooth hyperbreaks to capture nonmonotonic scaling behaviors.

3.1.1 Power-Law Formulations

Cross-Modal Variance in Power-Law Exponents.

Significant heterogeneity characterizes the data scaling exponents across distinct domains, ranging from near-zero sensitivity in remote sensing to superlinear requirements for code generation [127, 228]. While remote sensing operates within a data-limited regime with an exponent approximating zero point zero three, code generation exhibits a markedly steeper loss-data slope driven by high information density and syntactic repetitiveness. Protein language models further deviate from standard natural language trajectories by converging to a universal loss plateau near two point four that acts as a hard lower bound independent of parameter count [187]. Time-series foundation models maintain consistent power-law adherence across both in-distribution and out-of-distribution settings, yet specific architectural modifications, such as patch embeddings or alternative head designs, can suppress scalability slopes for generalization tasks [49, 239].

Finite-Size Effects and Low-Regime Breakdowns.

Standard power-law formulations exhibit a discontinuous breakdown in low-compute regimes, failing to predict loss accurately below specific FLOP thresholds even when vocabulary constraints are applied [46]. To address these deviations, estimator functions incorporating sigmoid-like transitions between random-guessing baselines and asymptotic power laws significantly reduce extrapolation errors compared to pure interpolation fits [4]. Furthermore, evidence suggests that such scaling laws emerge generically from optimization landscape properties rather than data distribution statistics, allowing monotonic exponent evolution across tunable complexity levels without inherent data power laws [11].

Embedding Parameters and Exponent Sensitivity.

The inclusion or exclusion of embedding weights significantly alters the fitted data scaling exponents, shifting the slope values that characterize loss reduction relative to dataset size. When embeddings are counted within the total parameter budget, the resulting power-law coefficients often underestimate the efficiency of non-embedding layers, whereas isolating transformer blocks yields steeper exponents that more accurately reflect the compute-cost of attention mechanisms. This sensitivity implies that reported scaling rates vary systematically based on whether the parameter count N encompasses static vocabulary projections or focuses solely on deep network weights, necessitating consistent accounting conventions to ensure valid cross-model comparisons [31].

Variance Sources and Correction Methodologies.

Fixing power-law exponents to universal constants derived from generic architectural mechanisms, such as Softmax nonlinearity and representational superposition, eliminates the ill-conditioned correlation between coefficients and exponents to stabilize loss extrapolation across diverse training configurations [119, 220]. This approach grounds the scaling behavior in a shared-parameter formulation where the Hessian spectral dimension dictates a specific universality class, thereby resolving parameter identifiability issues inherent in free-form fitting. Introducing multiplicative efficiency factors for model size and data volume further isolates optimizer-specific effects, effectively decoupling algorithmic improvements from structural scaling trends [220]. By establishing an AdamW reference baseline, these factors reduce mean squared extrapolation error through convex quadratic proxies that decompose optimization error from data efficiency dominance. Augmenting standard scaling laws with a dimensionless quality parameter bounded between zero and one corrects variance arising from corpus heterogeneity, revealing that effective sample size scales sublinearly with respect to data purity metrics [204]. The resulting relation defines effective data as the product of total data and the quality parameter raised to a power strictly less than one, quantifying the impact of corruption rates and synthetic noise injection while adhering to information-theoretic bounds on learnable content.

3.1.2 IsoFLOP Analysis

Curvature Dynamics and Budget-Dependent Vertex Shifts.

By fixing total compute budgets, IsoFLOP analysis constructs parabolic loss surfaces where the unique vertex identifies the optimal allocation between model parameters and training tokens [76]. This optimal vertex does not remain static; rather, its location shifts dynamically according to task-specific skill requirements, with knowledge-intensive domains favoring larger model capacities while code generation tasks prioritize increased data volume [172]. Distinct curvature dynamics further emerge in modalities with lower information density, such as discrete audio tokens, where optimal data scales significantly faster than model size compared to text-only baselines [135]. Consequently, the scaling exponents become asymmetric across modalities, reflecting how varying token information densities alter the trade-off surface under constant FLOP constraints.

Systematic Biases from Off-Center Sampling and Asymmetric Surfaces.

Fitting parabolic approximations to inherently asymmetric loss surfaces introduces systematic biases in vertex localization, causing

optimal allocation intercepts to drift independently of the compute budget [41]. These artifacts are exacerbated when training runs utilize off-center sampling grids, which distort both scaling exponents and intercepts, ultimately resulting in significant dead-weight compute loss at frontier scales compared to methods that isolate nonlinear parameters. Discrepancies in previously reported scaling exponents often stem from such confounding factors—including fixed warmup durations and inconsistent parameter counting conventions—rather than inherent flaws in the compute-optimal hypothesis, as corrected analyses resolving these variables align closely with theoretical predictions [167].

Geometric Distortion from Counting Conventions. Defining model size within a fixed compute budget fundamentally alters the geometric curvature of the resulting IsoFLOP loss surface rather than merely shifting its vertex. When parameter counting conventions vary, such as including or excluding the final output projection layer, the functional form of the parabolic fit degrades because the underlying relationship between compute and capacity becomes non-uniform across the sampling grid [167]. This inconsistency introduces structural noise into the surface topology, reducing the fidelity of quadratic approximations used to identify optimal regimes. In sparse architectures, these definition-dependent distortions are amplified, as the mismatch between total parameters and active FLOPs creates irregularities in the 3D loss landscape that obscure the true trade-off between capacity and inference efficiency [1]. Consequently, the reliability of the parabolic model itself depends on maintaining strict consistency in how architectural components contribute to the compute constraint.

Temporal Constraints and Hardware Bottlenecks. When wall-clock time replaces theoretical FLOP counts as the primary constraint, the optimal model size scaling exponent shifts from approximately 0.50 to 0.60 due to non-linear throughput variations driven by hardware bottlenecks [84, 118]. Memory bandwidth limitations and data transfer overheads serve as superior predictors for training duration compared to raw compute estimates, with runtime models based on memory copy features achieving an R-squared value of 0.74 in specific empirical settings. These physical constraints necessitate architectural adaptations toward wider multi-layer perceptrons and reduced layer depths to maximize utilization under single-TPU or GPU memory bandwidth limits. Consequently, the resulting loss surfaces exhibit a transition where short-duration regimes remain dominated by compute bottlenecks while extended budgets trig-

ger early data overfitting, a phenomenon observed across training windows ranging from five minutes to twenty-four hours [118].

Parabolic Deviations in Non-Standard Architectures. Standard quadratic approximations of IsoFLOP curves frequently fail when applied to architectures that decouple sequence mixing costs from parameter counts. In xLSTM models, the adoption of linear time-complexity mLSTM cells removes the quadratic penalty associated with expanding context windows, causing optimal model sizes to remain stable rather than shrink as sequence length increases [12]. The introduction of sparsity further complicates this topology; in Mixture-of-Experts systems, the interaction between total capacity and active parameter fractions transforms the two-dimensional loss parabola into a three-dimensional polynomial surface where optimal operating points depend on multiplicative sparsity terms [1]. Such geometric irregularities extend to multi-modal domains, where audio scaling frameworks exhibit sparse and discontinuous Pareto frontiers driven by resolution subsampling effects that defy smooth quadratic fitting [2]. These structural divergences indicate that parabolic assumptions hold only under strict homogeneity of compute mechanisms and break down when architectural axes introduce non-monotonic trade-offs.

3.1.3 Compute-Optimal Frontiers

Dynamic Token-to-Parameter Regimes. Foundational analyses establish a balanced scaling trajectory where model parameters and training tokens increase in equal proportion, yielding an exponent near 0.5 to minimize loss under fixed FLOP constraints [76]. Incorporating lifetime inference demand into the optimization objective fundamentally shifts this frontier, favoring significantly smaller architectures trained on data volumes where the token-to-parameter ratio reaches ten thousand [177]. Empirical validation at the billion-parameter scale confirms that exceeding standard recommendations through extended training on trillions of tokens yields competitive performance via multi-stage curricula [248]. Specialized learning rate schedulers further enable continuous training regimes that identify compute-optimal ratios substantially higher than initial theoretical predictions, often approaching one hundred ninety-two to one [80]. Sustaining progress along these evolving frontiers ultimately requires decoupling logical compute from physical resource burdens by integrating time-indexed efficiency doubling rates to mitigate diminishing returns [123].

Economic Regime Trade-offs. Microeconomic op-

timization frameworks formalize distinct expenditure scaling laws where training costs grow sub-quadratically with hardware efficiency in compute-bound regimes but quadratically with data availability in data-bound scenarios [68]. Unified architectural cost models validate that economic viability at extreme scales relies on predicting all context positions simultaneously rather than solely the final token to reduce total FLOP requirements, integrating decoder-only decomposition and Mixture-of-Experts decoupling [168]. Empirical efficiency benchmarks for small-scale models demonstrate that maximizing GPU memory utilization via standard parallelization strategies often yields suboptimal cost-per-token ratios compared to attention-optimized data-bound approaches, particularly when measuring Loss/Dollar metrics on modern accelerators [7].

Hardware-Constrained Scaling. When wall-clock time replaces floating-point operations as the primary constraint, the optimal model size scaling exponent increases to approximately 0.60, a shift driven by sublinear throughput degradation where tau scales proportional to N to the power of negative point eight [118]. This temporal bottleneck generates dual convex loss surfaces distinct from standard FLOP-constrained regimes, as validated empirically on consumer hardware such as the RTX 4090. Analytical performance models further demonstrate that optimal parallelism strategies must adapt dynamically to interconnect domain sizes and memory capacity limits, particularly within NVSwitch topologies where collective communication latency dictates iteration time minimization [203]. In reinforcement learning contexts, arithmetic intensity acts as a confounding variable that distorts parameter-based scaling laws, revealing an intrinsic performance frontier where the optimal size exponent fluctuates between 0.40 and 0.65 depending on task horizon length [75].

Divergence Between Pretraining and Fine-Tuning Optimal Frontiers. Extending the optimization horizon to include test-time sampling costs fundamentally alters the compute-optimal configuration, shifting the regime toward smaller, overtrained architectures that maximize inference efficiency under a joint life-cycle budget [173]. In multimodal contexts, this trade-off manifests differently; incorporating vision encoder compute within fixed constraints necessitates reducing language model parameters to accommodate higher visual token density and frame counts [221]. The dynamics of adaptation further complicate these frontiers, as fine-tuning trajectories exhibit a distinct pre-power phase driven by transferred knowledge that requires rectified scaling laws with pre-learned data parameters to identify optimal stopping points in

low-data regimes [109]. Finally, decomposing total tokens into dataset volume defined by example count and average length reveals that larger models achieve superior normalized token efficiency under fixed compute constraints compared to raw token counting methods [34, 96].

Heterogeneity-Induced Frontier Shifts. When data distributions exhibit high entropy, measured via compression ratios, the compute-optimal allocation shifts decisively toward larger dataset sizes rather than increased model parameters [158]. This reallocation interacts with architectural capacity thresholds, where under-parameterized models require diverse mixed corpora to mitigate underfitting, whereas larger architectures achieve superior sample efficiency on linguistically rich, single-domain texts [237]. Aggregate loss minimization often obscures these dynamics because distinct downstream skills, such as knowledge retrieval versus code generation, follow divergent scaling trajectories that can vary optimal parameter counts by up to 50% within a unified training run [172]. Furthermore, modality-specific information density disparities necessitate non-universal scaling exponents; for instance, discrete audio tokens demand data volume growth rates significantly exceeding those of text to reach equivalent performance saturation [135].

3.1.4 Extrapolation Techniques

Fitting Window Selection and Early-Stage Extrapolation Reliability. Excluding early training phases characterized by high noise or pre-power dynamics significantly reduces fitting error when extrapolating to larger scales, particularly when discarding token counts from initial stages until loss slope stabilization occurs [35, 109]. Utilizing intermediate checkpoints across the full training trajectory yields superior predictive accuracy compared to relying solely on final convergence losses, as dynamic hyperbolic fitting captures token-position specific parameters through temporal loss decomposition [35, 234]. Valid scaling trends in fine-tuning regimes further require independent hyperparameter optimization at each scale to prevent masking true performance gaps [85]. Iterative subsampling algorithms enable robust extrapolation by identifying the precise onset of power-law behavior before committing to full-dataset training, with methods like Accept then Stop (AtS) leveraging log-log space linearity on halved dataset subsets to determine optimal window boundaries [109].

Statistical Robustness and Confidence Interval Estimation. Theoretical analysis identifies the coefficient of variation as an architecture-invariant met-

ric that remains stable across scaling regimes, facilitating uncertainty quantification without requiring large-scale ensembles [51]. To rigorously bound extrapolated performance, hierarchical bootstrap procedures applied to log-log least-squares regressions generate confidence intervals that quantify the reliability of small-scale fits prior to deployment [85]. Furthermore, variance-based metrics derived from intermediate checkpoint fluctuations, such as the SD-10 statistic, predict task-specific extrapolation success more accurately than final loss values by capturing dynamics in the overtrained regime that decouple from sigmoidal accuracy mappings [14].

Validation Metrics: Prioritizing Extrapolation Error over Interpolation Fit. Minimizing interpolation error on small-scale data frequently fails to guarantee accurate large-scale predictions, necessitating objective functions that explicitly penalize extrapolation deviation rather than global curve fidelity [4]. Approaches employing square-log loss or relative error penalization within block coordinate descent frameworks effectively handle non-asymptotic regimes, preventing the divergence observed in standard estimators when projected beyond the training horizon. Reliable forecasting of asymptotic performance plateaus becomes achievable using minimal early-stage samples when fitting procedures target specific downstream accuracy thresholds, such as ROC-AUC at sample size n , instead of attempting to match the entire scaling trajectory [150]. This sample-efficient strategy allows for the precise determination of critical sample sizes through early slope prediction and linear probe fine-tuning. Furthermore, replacing scalar aggregate losses with full token-level probability distributions enables zero-shot extrapolation of complex, non-monotonic behaviors that rigid parametric families cannot capture [78]. By utilizing strided 1D CNN encoders optimized via pinball loss to predict quantiles spanning lower to upper levels, these methods preserve distributional signals essential for modeling irregular scaling dynamics.

Overfitting Risks in Noisy Small-Scale Regimes. Procedural under-specification in optimization and fitting choices induces severe fragility, where minor variations in learning rate schedules or parameter definitions yield divergent asymptotic predictions from identical small-scale data [104]. This sensitivity extends to the specific power-law form selected, the impact of initialization grids, and inconsistencies in reporting embedding parameter counts, all of which contribute to significant deviations in projected final loss values. Furthermore, structural asymmetries in loss surfaces cause standard parametric approximations to systematically bias intercepts and exponents

when fitted on off-center small-scale sampling grids [41]. Such off-center sampling distorts the perceived curvature of the scaling trajectory, leading to substantial prediction errors when these biased projections guide large-scale training decisions due to overfitting artifacts rather than genuine convergence properties.

3.1.5 Parameter Counting Conventions

Bias Induction via Embedding Exclusion. Excluding embedding parameters from the total count in small-scale regimes artificially inflates the observed scaling exponent, creating a discrepancy between local and global power laws [163]. While multiplicative ambiguities in parameter definitions leave optimal token-to-parameter ratios invariant, additive biases introduce spurious slope dependencies that distort compute-optimal strategies [182]. The functional form relating total to non-embedding parameters reveals that small-model curvature in the corrected loss surface replicates the higher exponents previously attributed to fundamental scaling differences [163]. Empirical re-fitting demonstrates that standard formula counts yield flatter optimal data trends compared to reported values, confirming that reporting inconsistencies rather than architectural limits drive observed deviations [182].

Distortion of Compute-Optimal Frontiers. Injecting frozen random parameters or employing structured transforms to artificially inflate model dimensions fails to alter loss-compute trajectories, confirming that scaling dynamics depend strictly on trainable parameter counts rather than total memory footprint [31]. Discrepancies in reported scaling exponents frequently originate from inconsistent FLOP accounting practices, particularly the exclusion of last-layer operations, which distorts the derivation of compute-optimal frontiers and inflates previously established power-law coefficients [167]. Standardizing parameter definitions to encompass all computational components reconciles these divergent predictions, demonstrating that prior conflicts arose from methodological omissions regarding additive biases and reporting inconsistencies rather than fundamental architectural limits [31, 167].

Standardization Protocols for Cross-Study Comparability. Systematic variation in parameter counting definitions, particularly regarding the inclusion or exclusion of embedding layers, induces significant divergence in derived optimal token-to-parameter ratios [104]. This ambiguity creates fitting fragility where procedural under-reporting leads to substantial deviations in the ratio of data tokens to parameters and sensitivity across optimization grids, necessitat-

ing explicit reporting protocols to ensure cross-study reproducibility. To address cases where metadata remains withheld, constructing high-dimensional accuracy profile vectors from next-token prediction tasks enables the statistical isolation of total parameter counts [154]. Such frameworks utilize latent size index extraction via PCA dimensionality reduction and blocked sign-permutation tests to verify model sizes and establish conservative lower bound estimates for closed-weight systems.

3.1.6 Broken Power Laws

Mechanistic Origins of Phase Transitions. Discontinuous capability gains arise from the sequential saturation of orthogonal skill bases governed by data distribution frequencies rather than smooth gradient updates [148]. This process follows a multilinear model where independent basis functions exhibit sigmoidal strength evolution determined by a frequency exponent, creating distinct stage-like training dynamics. Inverse depth scaling manifests differently as an ensemble averaging mechanism, wherein redundant layers reduce error variance instead of performing hierarchical feature composition [120]. Decomposed loss terms and angular update analysis reveal that independent weight regimes yield a depth exponent near unity, indicating that procedural assembly failure drives this specific non-monotonicity. Furthermore, non-monotonic double descent curves result from a scale-time equivalence where small parameter counts induce effective under-training and premature noise acquisition [20]. Random subspace analysis demonstrates that the product of parameters and tokens governs spectral decomposition, causing early noise dominance to produce U-shaped generalization error profiles.

Functional Forms for Discontinuous Regimes.

Standard scaling equations generalize to smooth broken power laws by introducing tunable sharpness parameters that model non-monotonic double descent and inflection points on linear-linear plots [27]. Multivariate extensions incorporate additive symmetries and reciprocal operations to explicitly capture oppositional forces from overfitting and suboptimal hyperparameters across multiple scaling dimensions [28]. Differential analysis further isolates interaction terms between model size and data volume, utilizing stretched-exponential parameterization to account for data scaling rates that vary systematically with parameter count [101]. These formulations collectively characterize phase transitions where simple monotonic trends fail to describe empirical loss surfaces.

Predictive Utility for Capability Transitions and Downstream Tasks.

Rectified scaling formulations integrate pre-learned data size parameters to distinguish pre-power phase transitions from stable power-law regimes, enabling accurate downstream performance extrapolation from sparse fine-tuning curves [109]. Task-stratified frameworks decompose aggregate accuracy into orthogonal capability dimensions where quantization-induced phase transitions impair reasoning chains more severely than memorization or application tasks [253]. Vocabulary filtration strategies shift the compute-optimal breakdown threshold, allowing small-scale models to maintain strong upstream-downstream correlations until hitting a distinct FLOP ceiling below which standard power laws fail discontinuously [46].

Regime Shifts via Resource Imbalance. When the Over-Training Ratio surpasses a specific threshold, scaling exponents stabilize around a consistent mean value, marking a distinct transition from parameter-bound to data-bound inefficiency driven by high information density [32]. This phase change is quantified through dual-perspective density metrics that track intra-cluster concentration and inter-cluster separation alongside decay factors, revealing diminishing marginal gains as the system enters the over-trained regime. To preserve mathematical continuity across such transitions, unified frameworks incorporate sparsity-adjusted entropy terms scaled by a power of the complement of sparsity and linear penalties proportional to sparsity that prevent discontinuous loss predictions during compute allocation changes [77]. These formulations enforce a zero-sparsity continuity constraint while modifying the irreducible loss term, ensuring that active parameter normalization via the complement of sparsity maintains an invariant data-dependent component across both dense and sparse architectures.

3.1.7 Discussion

Theoretical power-law universality frequently fractures under empirical scrutiny, where domain-specific data constraints and architectural bottlenecks induce significant deviations from predicted scaling trajectories [44, 144, 148, 228]. Observed phenomena such as flat parameter scaling slopes ($b \approx 0$) in data-limited regimes and abrupt feature learning phase transitions suggest that multilinear skill-basis orthogonality and excess risk spectral bulk dynamics often override simple monotonic decay assumptions. Furthermore, compute-optimal frontiers remain highly sensitive to hyperparameter calibration and methodological artifacts, implying that apparent violations of scaling laws frequently stem from suboptimal training config-

urations rather than fundamental theoretical failures [12, 90, 167, 238]. Specific issues including fixed warmup duration artifacts, AdamW β_2 sensitivity, and the linearity of effective learning rates can distort IsoFLOP parabolic loss surfaces, thereby obscuring the underlying linear log-compute trade-offs. Reliable extrapolation and transfer viability ultimately depend on resolving inconsistent parameter counting conventions, particularly regarding embedding exclusion discrepancies and task-specific layer relevance, while accounting for non-monotonic behaviors such as double descent or abrupt capability emergence [19, 43, 73]. Without standardizing these metrics and modeling memorization Poisson processes alongside bias amplification phase changes, estimates of the effective data transfer exponent (α) will continue to exhibit unacceptable variance across model families.

3.2 Architectural Efficiency and Sparsity Scaling

This section examines how architectural innovations decouple parameter counts from active compute, fundamentally altering standard scaling relationships. We first analyze the distinct scaling behaviors of sparse Mixture-of-Experts models in Sec. 3.2.1, followed by an investigation into structured pruning strategies and their interaction with training duration in Sec. 3.2.2. The discussion then extends to linear-complexity mechanisms that address context length constraints in Sec. 3.2.3, alongside a critical evaluation of depth-width trade-offs for compute-constrained regimes in Sec. 3.2.4. Finally, we assess the impact of vocabulary expansion on overall performance in Sec. 3.2.5 and explore the efficiency gains provided by multi-exit granularity in Sec. 3.2.6.

3.2.1 Mixture-of-Experts Scaling

Optimal Expert Count and Parameter Scaling. Under fixed compute budgets, optimal Mixture-of-Experts configurations systematically increase total parameters while reducing active parameter counts to minimize pretraining loss [1, 77]. IsoFLOP surface analyses reveal an inverse relationship between total parameter count N and the active fraction, often modeled via sparsity-adjusted entropy terms or linear penalties that favor extreme sparsity as capacity grows. Unified scaling frameworks demonstrate that such architectures achieve superior data efficiency compared to dense counterparts, enabling larger model scales for identical token budgets through optimized resource allocation toward total parameter count N rather than data volume D [214, 222]. This gain relies on expert granularity exhibiting a non-

linear modulation on scaling trajectories; empirical results indicate that the optimal number of experts generally increases with total parameter count, as gradient noise scales favorably within expanding sparse regimes before diminishing returns set in. However, downstream reasoning capabilities diverge from these pretraining loss trends, frequently favoring denser configurations despite the theoretical optimality of sparse models for next-token prediction [1, 214]. Evaluations highlight a perplexity-reasoning mismatch, suggesting that the activation ratio critical for minimizing loss may not align with the inference-time compute per token required for complex reasoning tasks.

Communication Overhead and Distributed System Bottlenecks. Scaling MoE architectures to massive parameter counts necessitates advanced parallelism strategies that explicitly optimize network topology to mitigate inter-node communication latency [48, 111]. These approaches employ 4D parallelism through tensor, context, pipeline, and data decomposition across large-scale clusters, such as those utilizing 24K GPUs, where network bandwidth saturation often dictates model-system co-design objectives rather than raw compute availability. Specialized distributed optimizers introduce non-trivial overheads due to full gradient gathering requirements for matrix orthogonalization, contrasting sharply with standard element-wise update mechanisms [115]. Such methods incur significant latency during complex iterative reconstruction steps within state partitioning frameworks, creating bottlenecks distinct from forward pass routing costs. Holistic efficiency frameworks consequently categorize sparsity and MoE implementations as system-level design choices where memory footprint and computational trade-offs are governed primarily by data movement constraints rather than FLOPs alone [111]. This perspective shifts optimization focus toward hardware-aware resource allocation and the mitigation of network congestion arising from dynamic expert routing, addressing the physical limits of data transfer without decoupling memory and compute demands.

Capacity Constraints and Load Balancing Regularization. Enforcing fixed expert capacity establishes a hard ceiling on the token-to-parameter ratio, where exceeding this threshold triggers routing saturation and immediate token dropping [126, 214]. To mitigate this, auxiliary load balancing losses act as a regularization mechanism that penalizes expert utilization variance, effectively relaxing strict capacity bounds to preserve gradient flow [214, 222]. This introduces a critical trade-off: while aggressive capacity limits maximize sparsity, they risk discarding

Table 2 Taxonomy of representative methods in **Architectural Efficiency and Sparsity Scaling**. This table presents a taxonomy of architectural efficiency methods, shows distinct sparsity mechanisms and mathematical scaling forms, and highlights how active compute decouples from total parameters.

Method	Venue	Sparsity Mechanism	Scaling Formulation	Highlight
Mixture-of-Experts Scaling (Sec. 3.2.1)				
Power-Law Scaling for MoE [222]	EMNLP’24	Expert Count Scaling	Unified Loss Equation	Derives unified scaling law linking loss to experts, revealing reduced gradient noise and higher data efficiency.
IsoFLOP Surface Analysis [1]	ICML’25	Sparse MoE	3D IsoFLOP Surface	Fits 3D IsoFLOP surface optimizing total parameters against sparsity under fixed compute constraints.
Joint MoE Scaling Laws [126]	ICML’25	Switch MoE	Logarithmic Expert Dependence	Models logarithmic expert count effects on exponents while integrating memory constraints for joint training-inference optimality.
EL [214]	arXiv’25	Granular MoE	Efficiency Leverage Law	Quantifies Efficiency Leverage via activation ratio and granularity, identifying optimal expert counts through log-polynomial relations.
Generalized Scaling Law [77]	arXiv’25	Active Parameter Scaling	Sparsity-Modified Chinchilla	Extends Chinchilla law with sparsity-dependent entropy and linear penalties to unify dense and sparse scaling behaviors.
Structured Pruning and Sparsity (Sec. 3.2.2)				
Joint Scaling Law [54]	ICLR’23	Magnitude Pruning	Multiplicative Capacity	Derives scaling law modeling sparsity as multiplicative capacity factor independent of data scaling terms.
IsoFLOP Surface Analysis [1]	ICML’25	Sparse MoE	3D IsoFLOP Surface	Formulates IsoFLOP surface optimizing total parameters and sparsity under fixed compute budgets.
LLM Compression Laws [185]	arXiv’25	Structured Pruning	Power-Law Recovery	Establishes power-law linking compression ratios to recovery data, deriving critical infeasibility thresholds.
Linear-Complexity Architectures (Sec. 3.2.3)				
Linear Complexity Scaling [191]	EMNLP’24	Linear Attention	Dual Power-Law	Derives distinct scaling exponents and identifies retrieval deficits from fixed hidden states in linear recurrence models.
MLIP Foundation Model [244]	Nature’25	Mixture Linear Experts	Data-Driven Distillation	Proposes Mixture of Linear Experts to replace hard-coded physical constraints with data-driven learning for scalability.
IsoFLOP Scaling Analysis [12]	arXiv’25	Linear Complexity LSTM	IsoFLOP Profile	Demonstrates xLSTM Pareto-dominance over Transformers via larger compute-optimal sizes and robust context-length scaling.
Depth-Width Trade-offs (Sec. 3.2.4)				
DeepNarrow Scaling [211]	arXiv’21	Depth-Width Trade-off	Pareto Efficiency Curve	Prioritizes depth over width to maximize Pareto efficiency and reduce parameters while maintaining downstream accuracy.
Bian et al. [17]	ICML’25	Aspect Ratio Tuning	Latency-Constrained Loss	Integrates aspect ratio into scaling laws to optimize width-depth configurations under strict inference latency constraints.
Gemstones Model Suite [143]	NeurIPS’25	Width-Depth Ratio	Hybrid Convex Hull	Derives explicit width-depth scaling terms via convex hull fitting to identify optimal aspect ratios across FLOP budgets.
Vocabulary Scaling (Sec. 3.2.5)				
Reduced-Vocab PT [46]	ACL’23	Vocabulary Reduction	Compute-Optimal Deviation	Filters vocabularies to enable pre-training, revealing power law breakdowns in low-compute regimes.
Vocab-Aware Scaling Laws [210]	NeurIPS’24	Tokenization Efficiency	IsoFLOPs Minimization	Derives optimal vocabulary scaling via unigram-normalized loss and derivative-based FLOPs minimization.
Zipfian Alignment Score [70]	EMNLP’25	Vocabulary Optimization	Zipfian Goodness-of-Fit	Selects optimal vocabulary by maximizing Zipfian linearity through iterative BPE expansion and R-squared monitoring.
Multi-Exit Granularity (Sec. 3.2.6)				
Familial Scaling Law [198]	arXiv’25	Early Exiting	Hierarchical Decomposition	Formulates unified scaling law $L(N,D,G)$ quantifying negligible granularity overhead and upstream-downstream interference.

high-value tokens unless counterbalanced by sufficient regularization strength, which directly modulates the effective scaling exponent by preventing the collapse of expert diversity [126, 222].

Divergent Memory Efficiency Profiles. By activat-

ing only sparse subnetworks during forward passes, MoE architectures decouple the static model footprint from dynamic compute requirements, substantially reducing the memory needed to store total parameters relative to dense equivalents with comparable capacity [126, 168]. This separation allows

the active parameter count to remain constant while the total parameter count scales independently, effectively lowering inference FLOPs without sacrificing model expressivity. Such divergence underpins the economic viability of billion-scale domain-specific models, where massive total capacity fits within static memory constraints while fixed top- k routing ensures stable wall-clock latency [168, 192]. Consequently, capital cost estimations favor sparse FFN substitutions that maintain constant inference costs even as large-scale domain-specific corpora drive requirements for larger total parameter counts.

3.2.2 Structured Pruning and Sparsity

Joint Scaling Formulations. Contemporary scaling laws reframe sparsity as a primary multiplicative factor acting on parameter capacity rather than an independent additive penalty, thereby preserving the functional form of data-dependent terms [54, 77]. This formulation introduces a normalization term based on the complement of the sparsity ratio raised to a specific exponent to scale active parameters, while maintaining a separate power-law relationship for data scaling to ensure performance predictions remain consistent across varying density levels. To enforce mathematical continuity at zero sparsity, unified frameworks adjust the irreducible loss entropy via an exponential factor dependent on sparsity and incorporate linear penalty terms proportional to the sparsity ratio, which prevent discontinuities in loss prediction and recover the dense baseline when sparsity vanishes [77]. Such adjustments yield smooth IsoFLOP curves that accurately reflect the trade-offs inherent in sparse training regimes. Furthermore, joint power-law derivations establish error-preserving invariants based on rational envelopes and saturating power-laws, demonstrating the interchangeability of architectural dimensions including depth, width, and pruning density while adhering to generalization bounds [174]. These invariants support the neural sampling conjecture by showing that equivalent generalization performance can be achieved through diverse combinations of structural compression and model expansion.

Temporal Dynamics of Sparsity Thresholds.

As training trajectories extend beyond standard compute-optimal data regimes, the optimal sparsity level exhibits a monotonic positive correlation with duration, shifting toward higher compression rates to mitigate overfitting on massive data volumes [54]. Under fixed FLOP constraints, IsoFLOP surface analysis reveals that loss-minimizing configurations necessitate an inverse trade-off between total parameter count and active parameters, effectively expanding

model size while increasing sparsity [1]. This divergence creates parabolic loss profiles where pretraining efficiency peaks at high sparsity, yet downstream reasoning performance often retains a preference for denser architectures despite the theoretical optimality of sparse scaling.

Hardware Realizability Constraints. Imposing rigid n -to- m connectivity patterns often yields diminished accuracy returns compared to unstructured approaches when targeting aggressive compression ratios, as the structural constraints limit representational capacity [54]. Even when algorithmic sparsity is achieved, theoretical speedups frequently fail to materialize because non-sparse operations, specifically layer normalization and attention mechanisms, dominate end-to-end execution latency on current software stacks [54, 235]. Bridging this gap necessitates co-design strategies that explicitly align sparsity patterns with the memory hierarchies and low-precision arithmetic capabilities of underlying accelerators rather than treating pruning as a purely algorithmic optimization.

Pruning Schedule and Recovery Efficiency.

When compression occurs after pretraining, performance degradation follows a power-law trajectory where increasing the pruning ratio eventually triggers an irreversible collapse regardless of the volume of recovery data employed [185]. This phenomenon establishes a hard feasibility boundary defined by a critical threshold, beyond which the quadratic increase in loss overwhelms any linear accuracy gains achievable through additional fine-tuning tokens. Conversely, integrating rank reduction iteratively during the training lifecycle circumvents this catastrophic failure mode by interleaving matrix merging steps with brief intervals of continual pretraining [65]. Such progressive low-rank decomposition preserves the original layer depth while enabling the derivation of multiple distinct model sizes from a single parent architecture, requiring only modest token budgets of approximately two hundred and fifty million per reduction step to maintain stability.

3.2.3 Linear-Complexity Architectures

Compute-Optimal Scaling and Parameter Efficiency.

Under fixed FLOP budgets, linear-complexity architectures follow distinct compute-optimal scaling trajectories where the optimal parameter count significantly exceeds that of quadratic attention models [12, 191]. This divergence arises because linear mechanisms avoid the sequence mixing overhead inherent to transformers, allowing larger model sizes to remain efficient without the memory-

bound constraints that force standard architectures to reduce parameters as context length grows. While these systems align with general power-law loss trends, their reliance on fixed-state compression introduces specific retrieval capacity deficits absent in full attention mechanisms [191]. The lack of a global token binding process restricts information recovery from long contexts, leading to measurable accuracy drops in tasks requiring precise needle-in-a-haystack retrieval compared to their quadratic counterparts.

Retrieval Capacity Deficits. When context lengths extend into extreme regimes, linear recurrence mechanisms encounter catastrophic retrieval degradation because fixed-dimensional hidden states cannot preserve distinct historical tokens for recomputation [191]. This architectural bottleneck prevents global token browsing, causing performance on precise retrieval tasks to collapse; for instance, accuracy drops to approximately 30 percent at a context length of 4.8K in needle-in-a-haystack evaluations. While state-space backbones such as S4 demonstrate proficiency in spectral filtering for continuous signals like ECG temporal dynamics, their memory capacity remains strictly bounded by static state dimensions rather than scaling with input duration [3]. Representational similarity analyses and linear probing confirm that these models operate within a finite memory horizon, fundamentally limiting their ability to match the dynamic retrieval capabilities of attention-based architectures in domain-specific sequential modeling.

Architectural Inductive Biases and Domain-Specific Data Efficiency. Substituting global attention with serialization strategies that encode geometric priors directly into input ordering enables linear-complexity backbones to achieve high data efficiency in irregular scientific domains [161]. Techniques such as hierarchical raster scans and 3D space-point serialization preserve spatial locality and trajectory coherence, allowing selective state spaces like Mamba2 to exploit these structural constraints without quadratic overhead. Shifting the burden of satisfying domain symmetries from hard-coded layers to data-driven loss objectives further facilitates scalable pre-training on diverse corpora [244]. To address instability arising from sparse detector data, domain-specific self-supervised objectives decoupled from strict autoregressive ordering provide necessary regularization [161]. Mechanisms including k-Next-Nearest-Neighbor prediction and event-difficulty loss re-weighting mitigate failure modes inherent to standard next-token tasks, ensuring stable representation learning even when sequential dependencies are weak or non-uniform.

Numerical Stability in Recurrent Unrolling. Extended sequence processing in billion-parameter linear architectures frequently induces numerical instability due to the cumulative propagation of rounding errors within recurrent states. Unlike attention mechanisms that reset context mixing at each layer, state-space models rely on long unrollings where small precision deviations in transition matrices can compound, leading to gradient explosion or vanishing signals in deep networks. To counteract these accumulation artifacts, careful initialization schemes and constrained spectral radii become essential for maintaining bounded hidden state norms without sacrificing representational capacity [161]. Specifically, enforcing bounded transition dynamics prevents the divergence of activation magnitudes during inference on ultra-long contexts, ensuring optimization trajectories remain well-conditioned even when sequential dependencies span millions of tokens.

3.2.4 Depth-Width Trade-offs

Compute-Optimal Aspect Ratio Shifts. Empirical evidence indicates that optimal architectural aspect ratios are not static but shift systematically as the training budget expands, generally favoring wider configurations at larger scales [5, 143]. This trend emerges from joint dimension optimization procedures, where fitting the convex hull of loss landscapes suggests scaling widths faster than depths, often adhering to learning rate schedules proportional to the product of width and the square root of depth alongside specific MLP dimension exponents. Domain-specific priors can invert this preference; for instance, weather modeling exhibits a distinct width bias driven by GNN memory-bandwidth bottlenecks and chaotic dynamics, rendering aggregate loss metrics derived from language tasks invalid for variable-specific RMSE minimization [242]. Furthermore, downstream transfer efficiency frequently necessitates deeper, narrower architectures that deviate from upstream pre-training optima, revealing Pareto inefficiencies in standard model shapes when evaluated on span-based masked language modeling tasks and SuperGLUE accuracy [211]. Such deviations highlight the tension between minimizing non-parallelizable computation during pre-training and mitigating vanishing gradients to preserve parameter reduction benefits during fine-tuning.

Stability Constraints in Deep Regimes. Theoretical analysis indicates that maintaining stability in deep residual networks necessitates scaling factors $\alpha \ll L$ that decay according to rates strictly determined by the Hölder regularity of underlying weight stochastic processes [138]. Failure to adhere to

these rapid decay requirements causes the Residual Neural Tangent Kernel to degenerate into constant functions, rendering infinite-depth architectures unlearnable regardless of parameter count [215]. Empirical observations in large language models suggest that depth scaling follows an inverse power law driven by ensemble averaging mechanisms rather than hierarchical procedural assembly [120]. Consequently, optimal network depth exhibits a non-monotonic performance ceiling where excessive layer counts induce convergence failures despite sufficient parameter budgets, often exacerbated by mixed-precision training defects and hardware memory constraints [190].

Variance Control via Outer Normalization. When the scaling factor associated with outer layer normalization nears a value of one, it becomes the dominant determinant of output variance, effectively suppressing the impact of inner layer width adjustments [241]. Asymptotic analysis indicates that under a fixed inner dimension, variance displays strict monotonicity relative to this outer parameter, rendering deeper architectural modifications negligible in the mean-field limit. Consequently, optimal aspect ratios for compute-constrained training must prioritize tuning these boundary scaling behaviors over expanding internal dimensionality once the outer factor approaches unity.

Hardware Parallelism and Memory Constraints. Training throughput in modern architectures is frequently governed by memory copy operations rather than raw floating-point counts, compelling a shift toward wider and shallower aspect ratios to minimize data movement overhead [17, 84]. Empirical analysis indicates that metrics tracking memory copies achieve an R-squared correlation of 0.74 with runtime, where expanding MLP width while reducing layer count mitigates latency variances observed up to 3.5 times in memory-bound regimes. Optimal parallelism strategies further adapt dynamically between 3D and 4D configurations based on interconnect topology, utilizing dual-bandwidth network modeling to assign GPU-to-NVSwitch domains and reduce collective communication latency within specific hardware clusters [203]. Such combinatorial searches account for High Bandwidth Memory capacity constraints, yielding performance predictions with an 11 percent error margin when deploying SUMMA tensor parallelism. In the context of deep graph neural networks, activation memory dominance forces a strict preference for width scaling over depth to avoid over-smoothing phenomena while enabling billion-parameter training through checkpointing techniques [99]. Integrating ZeRO optimizers in these scenarios reduces peak memory usage by 58 percent, addressing

the 76.9 percent activation memory share typical of EGNN backbones, albeit with a 133 percent increase in communication overhead.

Functional Divergence: Reasoning Depth versus Memorization Width. Under fixed parameter budgets, deep narrow architectures prioritize compositional reasoning capabilities over the statistical memorization favored by shallow wide configurations [82]. This structural preference manifests in tasks requiring multi-step inference, where increased layer count enhances performance on benchmarks such as GSM8k through shared embedding efficiency. However, expanding depth without proportional increases in optimization budgets or data volume induces severe performance degradation due to under-training in low-compute regimes [15]. Such compute-constrained scaling often results in convergence failure, characterized by elevated negative log-likelihood per FLOP and specific inefficiencies in positional encoding mechanisms like RoPE.

3.2.5 Vocabulary Scaling

Parametric Scaling Laws and Optimal Vocabulary Allocation. Conventional scaling frameworks often neglect the parameter cost of the embedding layer, resulting in suboptimal compute distribution where model depth and width expand while vocabulary size remains static or grows insufficiently [210]. This imbalance creates performance bottlenecks in both undertraining and overtraining regimes, as fixed FLOPs budgets are allocated to non-vocabulary parameters at the expense of token-character compression efficiency. To resolve this, a parametric loss function decomposes total capacity into distinct vocabulary and non-vocabulary components, enabling derivative-based minimization to identify the optimal allocation strategy. Empirical analysis utilizing IsoFLOPs power-law fitting reveals that the optimal vocabulary size scales as a power law with the compute budget, growing at a slower rate than the remaining model parameters with an exponent of approximately 0.83. Adhering to this allocation rule yields measurable accuracy gains on benchmarks such as ARC-Challenge across model sizes ranging from 33M to 3B parameters, although extrapolation beyond this range requires further validation.

Tokenization Granularity and Distributional Alignment. Conventional approaches often select fixed vocabulary sizes without a principled basis, yet token distributions adhering to power-law scaling suggest an optimal semantic granularity for downstream accuracy [70]. To address this misalignment, data-driven methods iteratively expand the vocabulary

via Byte Pair Encoding while monitoring the linearity of rank-frequency plots on log-log scales, halting expansion once the goodness-of-fit stagnates. This process yields a quantifiable metric based on the coefficient of determination, where peak model performance coincides precisely with maximum adherence to ideal Zipfian statistics. Such alignment provides a modality-agnostic validation signal, indicating that the quality of a tokenizer is intrinsically linked to its statistical fidelity rather than mere cardinality.

Vocabulary Constraints in Low-Compute and Edge Regimes. Restricting the tokenizer to domain-specific subsets, such as child-directed speech with a 21k-word limit, enables effective pre-training in sub-20M parameter regimes where standard vocabularies induce optimization failure [46]. This intervention addresses a discontinuous breakdown in compute-optimal scaling laws observed below the two point two quadrillion FLOP threshold, recovering the correlation between upstream perplexity and downstream GLUE performance that otherwise deviates from power law predictions. For edge-deployable architectures, shared input-output embedding weights decouple vocabulary magnitude from total capacity, sustaining a 256,000 token inventory within a 0.5B parameter budget by utilizing deep narrow topologies and Grouped Query Attention [82]. Such expansion implicitly extends the effective context length and minimizes unknown token probability without increasing the active parameter count, thereby prioritizing reasoning capabilities over memorization trade-offs during sequence length curriculum training.

3.2.6 Multi-Exit Granularity

Granularity Scaling Exponents. Theoretical frameworks for familial models extend traditional scaling laws by introducing granularity as a fundamental variable alongside parameter count and data volume, formalizing the relationship as a loss function dependent on parameters, data, and granularity [198]. Empirical analysis reveals a negligible granularity exponent, with a value of approximately 0.033 observed across IsoFLOP runs, indicating that supporting multiple exit points incurs minimal architectural overhead compared to single-exit baselines. This marginal cost confirms that the computational burden of maintaining decomposable branch networks scales sub-linearly relative to the number of deployed sub-models, preserving the efficiency of the shared backbone.

Inter-Branch Performance Interference. Theoretical disentanglement of scaling relations reveals that

upstream branch additions exert negligible performance penalties on downstream exits compared to the reverse configuration [198]. Empirical results indicate that incorporating additional downstream exit points incurs a linear loss penalty that persists even as total compute budgets increase. This directional nature of interference suggests that early exit placement primarily constrains later layers rather than being constrained by them, defining a specific hierarchy of dependency within the network architecture.

Compute Efficiency Leverage. Quantifying the computational advantage of multi-exit architectures reveals a consistent superiority over dense model ensembling baselines across varying FLOP regimes, as captured by the Efficiency Leverage metric where the value exceeds unity [198]. This gain amplifies significantly in low-compute settings, where extracting multiple outputs from a single forward pass delivers higher effective throughput than training independent models with equivalent parameter budgets. Consequently, the architecture maximizes utility per unit of compute by avoiding the redundancy inherent in maintaining separate models for different inference depths.

Mitigating Depth Saturation. By harvesting intermediate representations before the marginal utility of additional layers decays, multi-exit architectures counteract the performance saturation inherent in deep scaling [198]. This framework transforms the traditional compute-performance curve, enabling a single forward pass to yield multiple viable model instances and effectively flattening the efficiency drop-off associated with excessive depth. Consequently, computational resources shift from redundant deep layers to diverse exit points, sustaining performance growth where standard dense scaling would otherwise plateau under fixed budgets.

Convergence Instability in Joint Optimization. Simultaneous training of multiple exit points often induces convergence instability due to conflicting gradient signals propagated from shallow and deep layers within the shared backbone [198]. This interference arises because the weighted summation of exit-specific language modeling losses requires precise balancing; without it, early exits can dominate the update trajectory, thereby suppressing the learning of deeper representations.

3.2.7 Discussion

Achieving the theoretical efficiency promised by sparsity and linear-complexity mechanisms necessitates a rigorous co-design of model architecture and hardware infrastructure to effectively decouple total pa-

parameter count from active compute costs [111, 244]. Without such alignment, memory bandwidth bottlenecks inherent in sparse matrix multiplication and $O(N)$ state updates can negate the benefits of reduced active parameter ratios, rendering theoretical scaling laws impractical on current silicon. The long-term viability of these architectures further depends on trading rigid physical inductive biases for data-driven scaling strategies that maintain generalization across diverse domains, though this shift introduces challenges regarding transferability limits and distillation overhead [244]. Consequently, unified efficiency frameworks must jointly optimize numerical precision and structural sparsity to address the divergent memory and compute constraints distinguishing training from inference, particularly through advanced KV cache management and token reduction strategies [111]. These integrated approaches are essential to navigate the fidelity-compute trade-offs required for resource-constrained deployment while preserving the scaling advantages of large-scale sparse and linear models.

3.3 Data-Centric and Composition Scaling

This section shifts the analytical focus from raw token counts to the intrinsic properties of training data that govern scaling trajectories. Sec. 3.3.1 examines how weighting data sources by quality alters the substitution rates between high and low-quality tokens, while Sec. 3.3.2 investigates the impact of deduplication and repetition on memorization dynamics. The discussion extends to representation efficiency in Sec. 3.3.3, which analyzes how tokenizer choices modulate the relationship between token volume and semantic density. Furthermore, Sec. 3.3.4 and Sec. 3.3.5 explore scaling behaviors in domain adaptation and synthetic data generation respectively, addressing the sustainability of performance gains beyond natural corpora. Finally, Sec. 3.3.6 characterizes the limiting regimes where data scarcity dictates overtraining dynamics and defines the boundaries of effective reuse.

3.3.1 Quality-Weighted Mixtures

Effective Sample Size Formulation. Theoretical frameworks extend standard scaling laws by introducing a dimensionless quality parameter that modulates the data term via a sublinear exponent, often derived from information-theoretic principles or additive noise models [204]. This formulation redefines effective sample size as the product of raw token count and a quality-weighted power term, which quantifies the substitution rate between corpus volume and

fidelity while accounting for corruption robustness across strata defined by intrinsic information density and noise levels rather than topical domains [113, 204]. Alternative approaches model pretraining as cumulative information accumulation where high-quality sources provide higher density but exhibit scale-dependent diminishing returns under repetition, characterized by an exponential decay term within a unified power-law curve.

Entropy Proxies and Scale Invariance.

Compression-based metrics function as direct proxies for syntactic entropy, linearly mapping data compressibility to shifts in optimal parameter-data allocation frontiers [158]. When data density exhibits low inter-cluster separation, the resulting information redundancy stabilizes scaling exponents and triggers early diminishing returns driven by the intrinsic entropy limits of high-quality strata [32]. Empirical baselines further confirm that dataset quality rankings remain invariant across model scales, where high-fidelity English-dominant corpora outperform larger multilingual mixes despite lower total token counts [151].

Compute-Dependent Domain Allocation. Optimal mixture weights shift dynamically as a function of the total compute budget, necessitating joint modeling of domain ratios and model capacity [103, 195]. This dependency manifests as a compute-dependent optimal allocation parameter within the Joint Law formulation, where simplex-constrained optimization reveals that capacity-aware allocation drives non-proportional scaling across domains. Distinct modalities exhibit divergent substitution rates; specifically, the optimal data-to-parameter ratio for code grows exponentially with scale while remaining constant for natural language [127]. Such exponential growth in the ratio of data to parameters for code stems from a syntactic density gap and scale-dependent reversal effects, whereas cross-domain regularization noise limits similar expansion in text corpora. Furthermore, higher data quality fundamentally alters scaling exponents, favoring increased parameter scaling over data volume expansion compared to lower-quality sources [16].

Capacity-Dependent Complexity Thresholds. Architectural scale imposes a strict boundary on the utility of linguistically rich corpora, where models in the 18M to 44M parameter range frequently underfit high-quality data characterized by elevated Flesch Reading Ease, Automated Readability Index, and Gunning Fog scores [237]. In this low-capacity regime, performance on syntactic competence benchmarks inverts, favoring diverse mixtures of lower-complexity

Table 3 Taxonomy of representative methods in **Data-Centric and Composition Scaling**. This table presents a taxonomy of data-centric scaling methods, shows distinct optimization variables and targets, and highlights the shift from token counts to quality metrics.

Method	Venue	Scaling Variable	Optimization Target	Highlight
Quality-Weighted Mixtures (Sec. 3.3.1)				
Data Mixture Scaling Laws [195]	NeurIPS’25	Domain Weights	Loss Landscape	Derives joint scaling laws where optimal domain weights shift dynamically with model size and token counts.
CAMEL [103]	arXiv’26	Capacity Allocation	Benchmark Accuracy	Formulates capacity-aware scaling laws linking mixture ratios to model size via constrained optimization and hourglass sampling.
InfoLaw [113]	arXiv’26	Quality Buckets	Accumulated Information	Replaces compute metrics with quality-repetition aware information accumulation to unify scaling trajectories across diverse mixtures.
Deduplication and Repetition (Sec. 3.3.2)				
GP-MOLFORMER [175]	Digital’24	Prompt Vectors	Property Ranking	Quantifies training duplication bias impact on generative memorization and novelty decay.
InfoLaw [113]	arXiv’26	Quality Buckets	Accumulated Information	Models repetition-induced exponential decay to unify scaling trajectories via information metrics.
CEG [36]	arXiv’26	Repeat Count	Compute-Equivalent Loss	Derives compute-equivalent loss metrics revealing non-monotonic evaluation loss at intermediate repeat counts.
Tokenization Efficiency (Sec. 3.3.3)				
Embodied AI Scaling Law [164]	arXiv’24	Token Modality	Behavior Cloning	Tokenizer compression rates directly determine optimal model-to-dataset scaling coefficients.
Zipfian Alignment Score [70]	EMNLP’25	Vocabulary Size	Zipfian Fit	Maximizing Zipfian power-law adherence in token distributions optimizes downstream accuracy.
Information Capacity (IC) [243]	arXiv’25	Model Count	Information Capacity	Information capacity metrics integrate tokenizer efficiency to predict single-reference model performance.
Domain Adaptation Scaling (Sec. 3.3.4)				
Domain-Adapted Classifier [231]	Applied’24	Context Window	Template Selection	Linear correlation links language modeling loss to downstream classification performance during domain adaptation.
D-CPT Law [170]	NeurIPS’24	Mixture Ratio	Cross-Domain Loss	Parametric function integrates data mixture ratios into scaling laws to predict optimal corpus composition.
Data Mixture Scaling Laws [195]	NeurIPS’25	Domain Weights	Loss Landscape	Joint scaling law captures interactions between compute budgets and domain weights to derive optimal mixtures.
Knowledge Infusion Law [129]	EMNLP’25	Injection Frequency	Memory Collapse	Analytical scaling law predicts optimal knowledge infusion frequency before memory collapse occurs across model sizes.
DAPT [166]	arXiv’25	Token Intervals	Validation Loss	Empirical power-law exponents quantify data-efficiency frontiers trading financial specialization against general-domain forgetting.
Synthetic Data Scaling (Sec. 3.3.5)				
TabDPT [130]	NeurIPS’24	Context Rows	Retrieval Accuracy	Retrieval-based context alignment and self-supervised masking yield power-law scaling for tabular models.
CAUKER [233]	arXiv’25	Causal Kernels	Temporal Motifs	Composing Gaussian Processes with causal models reveals consistent data and model scaling laws.
PLUREL [94]	arXiv’26	Schema Complexity	Structural Fidelity	Hierarchical connectivity and conditional causal generation establish power-law scaling across database diversity.
Data-Constrained Regimes (Sec. 3.3.6)				
MLIP Foundation Model [244]	Nature’25	Dataset Diversity	Physics Compliance	Distillation transfers non-conservative teacher representations to efficient, physics-compliant student models for data-scarce domains.
DIVER-1 [67]	arXiv’25	Modal Domains	Reconstruction Error	Electrophysiology scaling favors smaller models trained longer over parameter growth under fixed compute and data constraints.
Akupara Dataset [228]	ICAIPR’25	Parameter Scale	Weak Supervision	Remote sensing exhibits data-limited scaling where model size increases yield negligible gains without commensurate data expansion.

tokens over exclusive training on curated sources like Project Gutenberg. Theoretical analysis using student-teacher perceptron frameworks indicates that prioritizing prototypicality and margin scores enables exponential error decay when model capacity aligns

with static data quality metrics, whereas mismatches adhere to slower power-law scaling [199].

3.3.2 Deduplication and Repetition

Non-monotonic Scaling and Compute Efficiency.

Contrary to the assumption that data reuse degrades performance monotonically, validation loss often exhibits a distinct peak at an intermediate repetition count before potentially stabilizing or declining, a phenomenon driven by the scaling of memorization capacity and block-diagonal noise covariance in mis-specified linear regression settings [36]. This non-linear trajectory implies that conventional parameter-centric metrics fail to capture the true efficiency costs of duplication, necessitating a Compute-Equivalent Loss framework to quantify degradation under fixed compute budgets where effective parameter reduction fallacies can obscure significant efficiency losses. To unify these dynamics across diverse mixture recipes, information accumulation models replace standard compute variables with quality-weighted repetition terms, utilizing exponential decay factors dependent on information density to describe power-law collapse during overtraining [113].

Scale-Dependent Memorization Velocity. As parameter counts increase, the number of exposure epochs required to reach verbatim memorization saturation decreases significantly, with larger architectures demonstrating an inverse scaling relationship between model size and data uniqueness requirements [216]. This accelerated convergence establishes a non-zero forgetting baseline where retention capacity grows monotonically with scale, preventing complete unlearning even during extended training phases. In parallel, the frequency of repeated sequences in training corpora acts as a primary determinant for generative memorization, directly correlating duplication rates with a reduction in output novelty fractions [175]. Strategic deduplication interventions decouple generation volume from this novelty decay, effectively shifting the exponential scaling law to sustain higher unique sample fractions across massive inference pools.

Empirical Validation of Deduplication Efficacy.

Rigorous application of string-level deduplication within macrodata refinement pipelines allows models trained on a single epoch of web-scale data to achieve zero-shot accuracy parity with systems trained on smaller, highly curated corpora [6]. This approach, exemplified by the RefinedWeb dataset, demonstrates that filtering for uniqueness can substitute for repetition-based learning strategies without compromising generalization capabilities. Conversely, removing duplicates does not universally guarantee performance gains or bias reduction across all evaluation axes. Analysis of The Pile dataset reveals

instances where deduplication yields null effects on overall utility, while specific stereotypical biases persist absent targeted counterfactual interventions [19]. These findings suggest that while deduplication optimizes data efficiency, it functions as a distinct mechanism from late-stage bias mitigation, requiring separate causal isolation strategies rather than serving as a panacea for representation harms.

3.3.3 Tokenization Efficiency

Compression-Driven Parameter Allocation. The optimal distribution of compute between model parameters and dataset size shifts fundamentally as tokenizer compression rates vary, directly altering the scaling coefficient such that optimal model size scales proportional to total compute raised to a specific power. Empirical analysis using GPT-2 causal transformers alongside VQGAN-encoded patches demonstrates that higher compression yields an optimal model size coefficient of 0.49, contrasting with the 0.66 observed in standard token regimes under infinite data conditions [164]. Decomposing total token counts into volume via example count and average sequence length further reveals that larger architectures achieve superior normalized token efficiency within fixed budgets. Evaluations across SMOLLM, QWEN, and Falcon backbones indicate that subsampling strategies favoring fewer long examples over many short ones maximize this efficiency metric, suggesting volume-aware scaling laws govern performance more accurately than raw token totals [96].

Semantic Granularity and Vocabulary Size Interactions with Model Depth.

Determining the optimal vocabulary size requires maximizing the linearity of token rank-frequency distributions rather than relying on arbitrary fixed counts, a process quantified by the Zipfian Alignment Score and R-squared goodness-of-fit on log-log scales [70]. Iterative BPE expansion continues until semantic granularity saturation occurs, ensuring the tokenizer captures statistical regularities without fragmenting coherent concepts. In deep architectures, adopting lower-base numeral systems further enhances data efficiency by increasing token frequency density, which mitigates carry extrapolation failures and extends length generalization limits compared to standard base 1000 representations [255]. Ultimately, this compression efficiency acts as the primary driver of total model information capacity, superseding parameter count variations across scales as evidenced by a Pearson correlation exceeding 0.98 between average text size per token and negative log-likelihood summation [243].

Modality-Specific Density and Architectural In-

variance. Discrete audio tokens inherently possess lower semantic information density than text, forcing an optimal scaling trajectory where dataset size must expand significantly faster than model parameters to reach acoustic fidelity saturation [135]. Despite such modality-specific constraints, the pretraining data distribution acts as the exclusive causal determinant for loss-to-loss scaling curves, rendering both architecture type and tokenizer choice invariant to the fundamental generalization trend across Transformer and State-Space Model variants [139]. This invariance extends to input sequence length, where larger encoder architectures enable aggressive token reduction through an inverse scaling relationship; performance remains stable with fewer input tokens provided specific semantic thresholds are met, though abrupt degradation occurs below critical limits such as eight tokens per image [105].

3.3.4 Domain Adaptation Scaling

Compute-Dependent Optimal Mixture Ratios. Optimal domain mixture coefficients function as dynamic variables that shift non-linearly with the total compute budget [170, 195]. This dependency arises from complex parameter-data interactions where the ideal balance between general and specialized knowledge evolves as model capacity and training duration increase. Predictive frameworks address this volatility by extrapolating shifting ratios from sub-billion parameter fitting, utilizing basin-hopping optimization within simplex-constrained loss landscapes to guide large-scale corpus composition without exhaustive grid searches [195]. Furthermore, explicit parametric formulations integrate these mixture ratios alongside model size and data volume into a unified scaling law dependent on model size, data volume, and mixture ratio, quantifying domain-specific learnability and convergence rates through power-law parameterization [170]. Such approaches enable the precise calculation of cross-domain transfer effects, ensuring that data allocation strategies adapt continuously to the specific constraints of the training regime.

Catastrophic Forgetting Dynamics and Memory Collapse. During domain-specific continual pretraining, specialized loss typically follows power-law decay while general knowledge retention remains stable within a bounded window up to a critical token threshold [166]. Beyond this limit, excessive knowledge infusion triggers memory collapse, where retention follows an inverted-U curve peaking at an optimal injection frequency defined by the ratio of specific constants before degrading as parametric saturation sets in [129]. Reusing overtrained base models ex-

acerbates this instability by introducing a logarithmic interaction term that reduces the scaling exponent for second-stage tokens, mechanistically linking performance saturation to elevated gradient norms in steep parameter regions [108]. Consequently, the theoretical scaling limit is defined by the intersection of these diminishing return curves and memory collapse boundaries, necessitating compute-aware truncation prior to the onset of irreversible performance degradation [108, 129, 166].

Divergence from Scratch Pretraining Laws and Efficiency Frontiers. Optimal compute allocation during domain-specific continual pretraining deviates fundamentally from training-from-scratch baselines, necessitating distinct power-law exponents for model size and data volume [34, 202]. IsoFLOPs profiling reveals that objective-dependent regimes, such as transfers from causal language modeling to masked language modeling, follow sublinear scaling trajectories where optimal model size and data volume scale as separate power functions of compute, differing markedly from general pretraining curves. This structural shift enables multiplicative efficiency gains; transferring a general base model to a specialized domain effectively reduces the compute required to reach target loss levels by an order of magnitude, often achieving parity with scratch-trained models using 7.7 times less computation [34, 202]. Such gains are quantified through Effectively Transferred Tokens (D_t), demonstrating that zero-shot adaptation requires orders of magnitude fewer examples while maintaining scale-dependent performance improvements. Furthermore, massive token counts during domain adaptation can redefine efficiency boundaries for inference-constrained regimes, allowing small-scale architectures to surpass standard scaling law recommendations [248]. For instance, a 1.1B parameter model trained on 3 trillion tokens with dynamic batch size cooldown achieves competitive performance against significantly larger baselines, challenging the assumption that parameter count must scale linearly with data volume in specialized contexts.

Domain Complexity and Convergence Thresholds. Scaling model parameters yields diminishing returns unless the training corpus operates within a compact, high-density semantic code space rather than unstructured text, as domain complexity imposes a hard convergence threshold [206]. In such regimes, mapping inputs to discretized identifier spaces and structured code sequences prevents overfitting while optimizing the sensitivity-specificity trade-off under compute constraints. Furthermore, language modeling loss during adaptation functions as a linear predictor for downstream discriminative per-

formance, establishing an efficiency frontier where extended token counts on smaller architectures match larger under-trained models [231]. This direct mapping between pretraining loss and classification error enables parameter-data substitution strategies that mitigate hallucination through targeted probability optimization and context window management.

3.3.5 Synthetic Data Scaling

Generative Fidelity as a Scaling Prerequisite.

Consistent power-law scaling in synthetic pretraining regimes depends primarily on the alignment between generator capability and output fidelity [94, 233]. Architectures that successfully encode high-fidelity structural and causal priors transform irregular real-world scaling trends into monotonic performance gains relative to model size and data diversity. Empirical evaluations indicate that maintaining this fidelity is a necessary condition for enhanced sample efficiency, as measured by improvements in zero-shot classification accuracy, coefficient of determination scores, and area under the receiver operating characteristic curve metrics. Without such precise correspondence between the generative process and underlying data constraints, scaling benefits degrade regardless of increased data volume.

Comparative Sample Efficiency. Pretraining on real data augmented with retrieval-based context alignment consistently demonstrates superior sample efficiency and faster convergence compared to purely synthetic generation pipelines, which often fail to replicate the full distributional complexity of natural corpora [130]. While synthetic pretraining functions as a scalable amplifier for building robust base models, these systems frequently exhibit a performance gap when transferred to downstream tasks without additional adaptation [94]. Empirical results indicate that although power-law scaling holds with respect to database diversity in synthetic settings, matching the performance of real-data baselines ultimately requires fine-tuning on limited high-quality real samples to bridge the inherent distribution shift.

Structural Priors for Niche Scarcity. In regimes where empirical distributions remain sparse, algorithmic priors enforce valid causal topologies and relational constraints to substitute missing samples rather than attempting to learn them from limited data [94, 233]. These generators inject specific discriminative signals absent in raw corpora, such as non-zero mean functions for temporal classification or exogenous patterns for relational regression, creating explicit supervisory cues that ensure causal-temporal coherence. By addressing this data void directly,

prior-driven synthesis converts the irregular performance plateaus typical of small-scale benchmarks into consistent monotonic scaling trajectories, evidenced by power-law consistency in UCR Archive zero-shot accuracy and RelBench coefficient of determination scores. Consequently, this approach enables the construction of foundational base models in verticals where privacy barriers or collection costs prevent the accumulation of tokens required for traditional scaling laws.

3.3.6 Data-Constrained Regimes

Loss Dynamics and Optimal Epoch Allocation.

When the Over-Training Ratio surpasses a critical threshold, fixed-data regimes transition from parameter-bound to data-bound inefficiency, causing loss scaling exponents to stabilize despite continued compute injection [32]. This saturation introduces a logarithmic interaction term during secondary training phases, where the scaling exponent for new tokens decays as the initial token count rises, ultimately hindering gradient norm progression and validation convergence [108]. Consequently, validation loss traces a U-shaped trajectory in which the optimal epoch count scales linearly with unique data volume yet inversely with model size, providing a predictive metric for overfitting onset [153]. Under such constraints, domains facing inherent scarcity maximize performance by prioritizing extended training durations on smaller architectures rather than pursuing parameter expansion, thereby avoiding the diminishing returns associated with large-model scaling under fixed budgets [67].

Architectural Divergence in Data-Starved Domains.

In specialized domains characterized by finite corpora, validation loss exhibits a flat scaling slope where the exponent approaches zero, indicating that orders-of-magnitude increases in model capacity yield zero marginal gain once the fixed token volume is exhausted [228, 249]. IsoFLOP analysis further delineates this boundary through parabolic efficiency curves, which identify a strict upper bound on optimal parameter count beyond which additional compute triggers immediate diminishing returns due to token starvation rather than improved representation learning. To mitigate these saturation effects, architectures targeting scientific data must actively reduce hard-coded inductive biases, as rigid physical constraints often prevent the effective utilization of diverse pre-training corpora necessary for scaling [244]. Replacing explicit conservation law enforcement with loss-based penalties or Mixture of Linear Experts allows models to bypass transferability limits inherent in physics-compliant distillation, thereby

extending the useful scaling regime even when data availability remains static.

Saturation Limits and Architectural Trade-offs.

In data-constrained regimes, the efficacy of multiple training epochs diminishes once the model exhausts the information content of the fixed corpus, causing loss scaling exponents to plateau regardless of additional compute injection [233]. This saturation point defines a hard boundary where further data reuse fails to reduce generalization error, necessitating a shift from dense parameter expansion to sparse architectural designs that maximize memory efficiency per unique token. Break-even analysis quantifies the threshold where continued pretraining on static data yields lower returns than alternative compute allocations, revealing that performance degradation accelerates when epoch counts exceed the optimal ratio dictated by dataset cardinality [88]. Consequently, avoiding overfitting in these starved contexts requires prioritizing architectures with reduced parameter density or increased sparsity, as standard dense scaling laws collapse when the volume of unique tokens cannot support the gradient updates required for larger models.

3.3.7 Discussion

Redefining optimal compute frontiers now requires prioritizing data complexity over sheer volume, where metrics such as gzip-compressibility and syntactic entropy dictate the parameter-data trade-off rather than simple token counts [135, 158, 164]. Strategic curation through deduplication and domain-specific weighting effectively mitigates memorization saturation, yet introduces non-linear retention collapse thresholds at specific infusion frequencies that challenge linear scaling assumptions [6, 19, 91, 129]. In data-constrained scientific domains, synthetic generation and retrieval-augmented pre-training restore monotonic scaling behaviors absent in small real-world corpora, although break-even points against classical methods remain highly task-dependent [83, 88, 130, 202, 233]. These dynamics collectively indicate that future scaling laws must account for the diminishing returns of web-scale data availability, shifting the primary optimization objective from acquiring more tokens to engineering higher-quality data compositions.

3.4 Training Dynamics and Optimization Scaling

This section examines the macroscopic behavior of optimization processes as model scale increases, focusing on the interplay between hyperparameter con-

figuration and convergence stability. We first analyze strategies for transferring optimal hyperparameters from proxy models to large-scale targets in Sec. 3.4.1, followed by an investigation into how specific optimizer choices influence scaling laws in Sec. 3.4.2. The discussion then extends to the mathematical modeling of loss evolution over training steps in Sec. 3.4.3, which provides a foundation for understanding batch size effects and gradient noise dynamics detailed in Sec. 3.4.4 and Sec. 3.4.6. Finally, Sec. 3.4.5 addresses the critical role of learning rate annealing schedules in determining final model performance under varying compute constraints. Together, these subsections establish a comprehensive framework for predicting and managing training dynamics across different scales.

3.4.1 Hyperparameter Transfer

Data-Dependent vs. Architecture-Dependent Scaling. Theoretical analysis of scalar linear networks indicates that universal power-law scaling fails without correcting for input-output correlation statistics, necessitating data-dependent exponents derived from conservation laws and finite-depth correction factors to ensure stable depth-wise transfer [250]. Empirical derivation across dense and sparse transformers further reveals a joint convex loss landscape where optimal learning rates follow a unified bivariate power law dependent on both parameter count and dataset size, often fitted via Ordinary Least Squares to yield scaling relations such as N to the power of negative 0.713 multiplied by D to the power of 0.307 while maintaining invariance to MoE sparsity [102]. Raw learning rate scaling frequently exhibits systematic upward curvature at large scales due to weight-norm transient dynamics inherent to AdamW, whereas defining the effective learning rate as a normalized weight space step size restores linearity and significantly improves out-of-distribution extrapolation accuracy [238].

Maximal Update Parameterization. Empirical validation on wafer-scale clusters demonstrates that maximal update parameterization reduces scaling variance by an order of magnitude compared to standard approaches, enabling direct hyperparameter transfer from 40M-parameter proxies to 2.7B-parameter targets without width-dependent tuning [47]. Theoretical analysis via Dynamical Mean Field Theory establishes that such mean-field parameterizations preserve optimal learning rates across widths, whereas Neural Tangent Kernel regimes necessitate adjustments that break transferability [21]. While depth-induced vanishing of forward-backward correlations restores the Gradient Independence Assumption in infinite-depth limits, structural separation within multi-layer

Table 4 Taxonomy of representative methods in **Training Dynamics and Optimization Scaling**. This table presents a taxonomy of optimization strategies and scaling formalisms, shows distinct mechanistic approaches and mathematical frameworks, and highlights the divergence between empirical fitting and theoretical derivation.

Method	Venue	Optimization Mechanism	Scaling Formalism	Highlight
Hyperparameter Transfer (Sec. 3.4.1)				
Corrected IsoFLOP Scaling [167]	NeurIPS’24	Hyperparameter Sweep	IsoFLOP Analysis	Corrects warmup and FLOP definitions to stabilize hyperparameter transfer across model scales.
Step Law [102]	arXiv’25	Convex Optimization	OLS Power-Laws	Derives joint power-laws for learning rate and batch size scaling with model and data.
AdamW Timescale Scaling [13]	NeurIPS’25	Timescale Fitting	Power-Law Trade-off	Models AdamW timescale to predict optimal weight decay and batch size via dataset ratios.
Optimizer Scaling (Sec. 3.4.2)				
Scaled Distributed Muon [115]	arXiv’25	Orthogonal Constraints	Empirical Laws	Distributed Muon algorithm scales Newton-Schulz iterations via ZeRO-1, yielding compute-optimal training with reduced FLOPs.
Scaling Collapse Analysis [169]	ICML’25	SGD Noise Dynamics	Universal Collapse	Quadratic SGD noise model predicts loss curve deformations and supercollapse phenomena induced by learning rate decay schedules.
Shared-Param Scaling Law [220]	arXiv’26	Unified Efficiency	Spectral Decomposition	Unified scaling law isolates optimizer-specific efficiency factors while sharing power-law exponents across different optimization methods.
Loss Trajectory Modeling (Sec. 3.4.3)				
DMFT [22]	ICML’24	Gradient Flow	DMFT Derivation	Derives exact Fourier response functions revealing asymmetric time-model scaling exponents and finite-width bottlenecks.
Temporal Scaling Law [234]	EMNLP’24	Token Granularity	Dynamic Hyperbolic	Fits token-position losses via dynamic hyperbolic laws with piecewise temporal parameters correlating to learning rate schedules.
Scaling Law w/ LR Ann [217]	NeurIPS’24	LR Annealing	Area-Based Modeling	Integrates annealing momentum into power-law scaling to predict full loss curves across unseen scheduler configurations.
Scaling Collapse Analysis [169]	ICML’25	SGD Noise Dynamics	Universal Collapse	Identifies supercollapse where normalized loss curves deviate below noise floors during learning rate decay phases.
NQS [100]	arXiv’26	Noisy Quadratic	Mechanistic Integration	Models batch size and LayerNorm effects on quadratic landscapes to derive closed-form loss trajectory expressions.
Batch Size Scaling (Sec. 3.4.4)				
AdamW Timescale Scaling [13]	NeurIPS’25	Timescale Fitting	Power-Law Trade-off	Derives critical batch size scaling as power laws in dataset size, independent of model parameters.
NQS [100]	arXiv’26	Noisy Quadratic	Mechanistic Integration	Integrates batch noise and LayerNorm effects into closed-form loss models for variable batch sizes.
Three-term Law (3TL) [184]	arXiv’26	Batch Optimization	Three-Term Power-Law	Formulates three-term loss law yielding optimal batch size scaling dependent on token budget.
Learning Rate Annealing (Sec. 3.4.5)				
Scaling Law w/ LR Ann [217]	NeurIPS’24	LR Annealing	Area-Based Modeling	Formulates annealing momentum term S_2 to predict full loss curves across unseen schedulers.
Scaling Collapse Analysis [169]	ICML’25	SGD Noise Dynamics	Universal Collapse	Derives scaling law where decay induces supercollapse by canceling shared stochastic noise.
Fwd-Momentum Scaling Law [223]	AAAI’25	Momentum Integration	Predictive Modeling	Integrates Adam-style momentum to stabilize annealing calculations and scale optimal ratios.
Gradient Noise Dynamics (Sec. 3.4.6)				
Exact Memorization Metric [216]	NeurIPS’22	Memorization Tracking	Exposure Curves	Larger models exhibit higher forgetting baselines, linking scale to memorization retention dynamics.

residual blocks still requires depth-aware corrections to prevent first-layer learning collapse [240].

Decoupling Batch Size Scaling from Model Dimensions. Theoretical formulations of the Three-term Law demonstrate that the critical batch size

depends primarily on the total token budget rather than parameter count, effectively decoupling optimal batch scaling from model dimensions [184]. Empirical analysis of AdamW dynamics corroborates this independence, revealing that both optimal and critical batch sizes follow power-law scaling with dataset

size regardless of model width, which enables direct transfer of data-dependent hyperparameters across scales [13]. Furthermore, distinct scaling exponents emerge for optimal batch size under fixed compute versus fixed data constraints, confirming that batch dimension growth is governed by resource allocation boundaries rather than architectural depth [193].

Failure Modes in Hyperparameter Transfer Across Architectures and Scales. Independent hyperparameter tuning per scale serves as a strict prerequisite for valid scaling law emergence, since fixed configurations induce systematic performance gaps that invalidate small-to-large extrapolation in finetuning regimes [85]. Relying on mean-relative-error metrics with coefficients of determination exceeding 0.95 reveals that task proximity dependency and encoder-only architecture limits frequently cause cross-objective model selection failures when proxy settings are rigidly applied. Furthermore, fixed warmup durations and suboptimal optimizer momentum coefficients at small scales create artificial inflation of scaling exponents, leading to erroneous compute-optimal predictions when transferred to larger models [167]. This distortion of IsoFLOP curves stems from sensitivity to the second moment decay rate in adaptive optimizers and the necessity of linear warmup scaling with parameter count, often resulting in significant discrepancies between Kaplan and Hoffmann exponent estimates when last layer FLOPs are excluded. Uniform spectral stabilization thresholds also fail across depths due to layer-dependent power law exponents, causing Newton-Schulz iteration breakdown in late layers when configurations derived from small proxies are applied to frontier scales [131]. Momentum singular value quantiles exhibit layer-specific exponents ranging from negative 0.25 to negative 0.96, meaning that burn-in stabilization strategies adequate for shallow networks underestimate the required step counts for deep architectures, ultimately degrading updates in the final MLP.

3.4.2 Optimizer Scaling

Second-Order Compute Dynamics. By approximating curvature information, specific optimizers reduce the total floating-point operations required to reach target loss levels compared to standard first-order baselines, effectively shifting the Pareto frontier for compute-optimal training [115, 220]. Empirical results indicate that such methods can achieve approximately 52% FLOP reduction while maintaining a data efficiency factor greater than one, suggesting a fundamental trade-off between per-step cost and convergence speed. Maintaining stability at scale

within these frameworks necessitates harmonizing update magnitudes across varying matrix dimensions through dimension-aware scaling factors, such as normalizing by the square root of the maximum dimension to prevent exploding weight norms [115]. Furthermore, reliable comparison of optimizer scaling behaviors demands unified power-law exponents coupled with optimizer-specific multiplicative efficiency factors rather than independent fits, thereby reducing extrapolation errors caused by ill-conditioned parameter correlations [220]. This approach links spectral dimensions directly to scaling coefficients, offering a more rigorous basis for predicting performance across model sizes.

Spectral Stability and Layer-Dependent Configuration. Post-burn-in dynamics reveal that momentum matrix spectra stabilize according to layer-specific power laws, necessitating non-uniform optimizer configurations across network depth [131]. Singular value quantiles decay following a power law with negative exponents that vary significantly between layers, indicating that a single global setting cannot capture the heterogeneous geometric landscape of deep architectures. Uniform second-order approximations frequently induce instability in deeper layers because rapidly decaying singular values violate orthonormalization thresholds required for stable Newton-Schulz iterations [131]. This precision mismatch causes update degradation along small singular value directions, particularly when extrapolating configurations from smaller proxies to frontier scales. Optimal scaling strategies therefore map spectral decay exponents, observed in the range from approximately negative one to negative zero point two five, directly to iterative refinement counts rather than applying static hyperparameters [131]. Such scheduling demands substantially higher computational steps for late layers compared to mid-layers, reflecting the divergent convergence requirements of MLP blocks versus projection matrices.

Distributed Communication Bottlenecks. As cluster counts expand without proportional bandwidth growth, inter-cluster synchronization latency increasingly dominates intra-cluster compute time, establishing a hard efficiency ceiling for distributed optimizers [69, 89]. This communication-computation imbalance degrades the K efficiency factor, where Ethernet latency bounds and nonblocking network requirements frequently force GPU idling during state partitioning. Linear loss penalties emerge directly from the frequency of local update steps, creating an unavoidable trade-off between synchronization overhead and convergence stability in multi-cluster configurations [69]. Specifically, periodic global synchronization al-

lows parameter discrepancy accumulation that varies the scaling exponent α , linking the local step count s to the loss formulation L as a function of K and N . Beyond algorithmic metrics, real-world carbon and time overheads expand by factors of two to five times ideal estimates because static power consumption and embodied emissions persist during communication-induced resource idleness [89]. The CarbonScaling framework highlights that utilization drops at one hundred trillion parameters drive these diminishing hardware returns, as the embodied carbon footprint remains fixed regardless of dynamic power fluctuations.

Optimizer-Induced Supercollapse. Acting as a control variate, specific learning rate decay functions can drive cross-model variance below the intrinsic seed noise floor during compute-optimal training, a phenomenon termed supercollapse [169]. This effect manifests when the scaling of variance reduction becomes proportional to the square root of the product between the learning rate and the complement of the normalized time ratio, indicating that stochastic noise cancellation becomes dominant as the training trajectory approaches its terminal phase. Consequently, the correlation between intermediate and final model states strengthens significantly, allowing the collapse quality to serve as a diagnostic for optimizer stability distinct from standard convergence metrics.

3.4.3 Loss Trajectory Modeling

Temporal Heterogeneity in Loss Decay Exponents. Loss decay trajectories exhibit distinct temporal regimes where early-stage dynamics remain independent of dataset volume before transitioning to data-dependent saturation [22, 42]. Aggregate loss curves often obscure position-specific convergence rates that follow dynamic hyperbolic distributions with evolving parameters correlated to scheduling strategies [234]. Theoretical analysis via dynamical mean field theory reveals that finite-width corrections and overfitting gaps accumulate gradually through repeated data reuse integrals rather than appearing instantaneously [22]. Uniform loss decrease rates across different token positions emerge only after an initial transient phase, enabling early-stage parameter extrapolation for full-training prediction once the system passes a specific separation point [234].

Granularity of Predictive Modeling. Relying exclusively on aggregate validation perplexity collapses distinct model scales onto a single performance trajectory, effectively obscuring individual token-level learning heterogeneities and masking divergent behaviors such as stagnation versus upward trends [232].

To resolve these hidden dynamics, mechanistic models must decompose test loss into stochastic optimization processes on quadratic surfaces characterized by power-law spectra, thereby capturing batch size and dimension interactions that aggregate scaling laws fail to represent [100]. This finer resolution reveals that convergence rates are not uniform across linguistic categories; specifically, unique identifiers like nouns and numerals drive faster exact match acquisition compared to the slower reduction suggested by global loss metrics [216].

Intrinsic Trajectory Curvature. Convex-derived asymptotic bounds characterize the limiting behavior of loss decay, where the curvature of the trajectory reflects the transition into weak convexity regimes independent of specific scheduling parameters [26]. Predictive frameworks decompose validation loss paths into cumulative forward updates and an explicit terminal decay component, isolating the deviation from power-law scaling that arises during late-stage saturation rather than attributing it to external cooling phases [217]. To maintain fidelity across diverse architectures, step-based tracking integrates bias-corrected first and second moments, modeling how intrinsic trajectory curvature scales with optimization state variables while filtering out artifacts introduced by batch size variations or mixture-of-experts sparsity patterns [223].

Early-Trajectory Extrapolation and Downstream Correlation. By parameterizing scaling laws through neural networks rather than static power functions, researchers can predict residual loss relative to a coarse baseline with high fidelity, effectively capturing non-linear hyperparameter interactions that rigid formulations overlook [246]. This approach utilizes a two-stage fine-tuning process with a mean squared error objective to generalize across out-of-distribution configurations, correcting for deviations from standard Chinchilla baselines. Conversely, longitudinal tracking reveals that performance correlations are not monotonic; larger models sometimes exhibit steeper accuracy declines on specific tasks as token counts increase, directly challenging assumptions of uniform capability growth [146]. Such inverse scaling phenomena, identified through z-scored regression analysis with false discovery rate corrections, indicate that parameter-token interactions can induce late-stage instability unique to certain downstream objectives.

Stochastic Signatures and Variance Collapse. During compute-optimal training, inter-model variance frequently undergoes a "supercollapse," dropping below the intra-model seed noise floor specifically while learning rate decay is active [169]. This phe-

nomenon adheres to a universal curve normalization where the deviation scales with the square root of the product between the learning rate and one minus the effective temperature parameter, acting as a statistical signature that suppresses cross-model fluctuations beyond the irreducible loss offset. High-granularity checkpointing further reveals that distinct behavioral traits, such as memorization and bias, follow stochastic emergence patterns independent of aggregate loss convergence rates [19]. Memorization events align with Poisson point processes exhibiting clear phase change thresholds and term frequency correlation onsets, whereas bias amplification dynamics remain reproducible under fixed data ordering despite counterfactual intervention windows. These observations suggest that neural scaling laws arise generically from observable trajectory properties rather than inherent data power-law structures, causing scaling exponents to evolve monotonically with dataset complexity [11].

3.4.4 Batch Size Scaling

Critical Batch Scaling Laws. The critical batch size exhibits a functional dependence on the loss value itself rather than model parameter counts, enabling the derivation of optimal time-compute trade-offs through implicit loss trajectory equations [201]. Empirical investigations characterize the optimal batch size as a power function of both compute budgets and data amounts, revealing distinct sub-linear growth exponents where optimal batch size scales with compute to the power of approximately 0.102 and with data to the power of roughly 0.264 under fixed resource constraints [193]. This scaling behavior extends to dataset dimensions where both critical and optimal batch sizes follow power laws proportional to data raised to the power of 0.5, independent of model width, thereby facilitating Pareto-efficient over-training strategies for time-constrained scenarios [13, 184]. Such relationships indicate that tokens-per-parameter ratios and AdamW timescales govern these limits more strictly than architectural depth or width.

Diminishing Returns and Communication Ceilings. Increasing batch sizes yields rapidly diminishing returns in gradient noise reduction, causing the time-compute efficiency curve to flatten once the critical batch size threshold is exceeded [92]. While larger batches theoretically reduce update steps, the resulting wall-clock gains saturate because the marginal improvement in convergence speed fails to offset the reduced sample efficiency per step [13]. This saturation point creates a hard ceiling on parallel scaling, distinct from optimal compute allocation, where further increases in local batch size provide negli-

gible loss reduction relative to the added computational cost. Furthermore, distributed architectures encounter strict efficiency limits where communication latency between clusters imposes a linear penalty that negates theoretical speedups from massive batch sizes without proportional bandwidth growth [69]. Consequently, the scaling trajectory is bounded not by model capacity or data volume, but by the point where inter-cluster synchronization time overwhelms the benefits of reduced gradient variance.

Learning Rate-Batch Coupling. Optimal learning rate scaling exhibits sub-linear growth relative to batch size increases, eventually saturating at a theoretical upper bound dictated by the gradient noise scale [193]. Empirical observations indicate that maintaining convergence requires adhering to this saturation limit rather than applying linear scaling rules, as exceeding the bound leads to optimization instability regardless of schedule adjustments [100]. Dynamic batch size doubling schedules function as an implicit regularization mechanism that mitigates loss spikes without requiring explicit learning rate decay phases, effectively navigating the coupling between batch magnitude and step size constraints [132]. This approach preserves training stability by respecting the noise-defined ceiling on learning rate magnitude, ensuring that batch expansions do not violate the empirical limits of convergent updates.

Generalization Gaps and Hardware Utilization Trade-offs. Optimal batch size configurations shift fundamentally when evaluation metrics diverge from local training proxies, particularly in two-stage architectures where in-batch sampled softmax fails to approximate full-catalogue ranking performance [55]. This metric-dependent scaling behavior arises because negative sampling constraints inherent to small batches create a representation gap that widens as the system transitions to memory-bound regimes. Beyond algorithmic fidelity, hardware utilization efficiency imposes a rigid ceiling on scaling, where increasing parallelism eventually yields diminishing returns due to fixed memory bandwidth and compute density limits. The resulting resource imbalance forces a trade-off where further batch expansion becomes counterproductive despite available theoretical throughput, as the marginal gain in gradient estimation accuracy cannot offset the stagnation in effective device utilization.

3.4.5 Learning Rate Annealing

Power-law scaling of annealing ratios. Empirical analysis indicates that optimal annealing ratios follow a consistent power-law decay relative to the total

training horizon T , requiring precise step estimation to avoid under-convergence in extended schedules [26, 223]. The peak learning rate acts as the primary determinant for this optimal annealing magnitude, adhering to a predictable scaling relationship between the initial value and the required decay extent. Theoretical validation confirms that qualified decay schedules achieve $O(1/\sqrt{T})$ convergence rates exclusively when the peak learning rate scales inversely with the square root of the iteration count, thereby satisfying necessary integral convergence conditions within the convex-dominance regime [26].

Decay Shape Dynamics. The curvature of cosine versus linear decay functions dictates the specific timing of convergence acceleration and the degree of intermediate stability during the annealing phase [80, 217]. Within the Warmup-Stable-Decay framework, linear phases facilitate sharp loss reductions even as gradient norms diminish, suggesting optimization dynamics that differ fundamentally from the smoother transitions of cosine schedules. Analytical models incorporating an annealing momentum term successfully predict full loss curves for unseen schedulers, thereby quantifying how decay shapes influence cumulative parameter updates [217]. Although both configurations often reach comparable final losses under optimal tuning, they exhibit significant divergence in intermediate stability and responsiveness to data injection across varying training horizons [80].

Variance Suppression via Annealing. By functioning as a control variate, learning rate decay suppresses cross-model variance through the correlation of intermediate optimization states with final convergence targets [169]. This mechanism drives a supercollapse regime where trajectory deviations fall below the intrinsic noise floor generated by random seed initialization. A derived scaling law quantifies this stability gain, linking the magnitude of collapse directly to the product of the learning rate and the remaining training fraction under a quadratic SGD noise model. Consequently, deviations from this universal collapse curve serve as a high-sensitivity diagnostic for identifying suboptimal data exponents or parameterization schemes before final loss metrics diverge.

Nonlinear Dynamics in Multi-Stage Decay. Aggressive decay schedules within multi-stage pipelines frequently induce transient weight-norm dynamics that distort raw learning rate scaling, thereby necessitating the use of effective learning rate metrics to accurately capture optimization progress [238]. Such transients compromise scale-invariance properties inherent to AdamW normalization, rendering standard log-linear extrapolation unreliable under

these nonlinear conditions. Conversely, fitting schedules along the data axis using effective learning rates significantly mitigates these distortions, reducing the Extra Compute Ratio required for stable hyperparameter transfer while improving out-of-distribution R-squared performance [238].

3.4.6 Gradient Noise Dynamics

Inverse Scaling of Gradient Noise Variance. Empirical investigations indicate that increasing parameter counts reduces the effective gradient noise variance, following an inverse scaling relationship with model capacity [216]. This suppression of stochastic fluctuations manifests as a monotonically decreasing variance baseline, where large models exhibit diminished volatility and converge to stable minima more rapidly than smaller counterparts. Consequently, the signal-to-noise ratio improves systematically as the number of parameters grows, modulating local optimization dynamics independent of specific data token characteristics or linguistic structures.

Curvature-Dependent Noise Regularization. Stochastic gradient fluctuations function as an implicit regularizer that inhibits convergence into narrow, high-curvature regions of the loss surface by introducing a sharpness penalty proportional to the Hessian spectrum. When the variance of injected noise exceeds the characteristic width of sharp minima, optimization trajectories naturally bias toward wide basins, a phenomenon describable through diffusion approximations of the stationary distribution. This dynamic establishes a critical stability threshold: excessive noise levels eject parameters from viable solution manifolds, causing convergence failure, whereas insufficient noise leads to entrapment in suboptimal sharp configurations due to under-exploration.

Stochastic Diffusion in Overparameterized Landscapes. Within regimes where model capacity significantly exceeds data complexity, the intrinsic stochasticity of mini-batch gradients acts as a mechanism for implicit regularization rather than mere optimization noise. This stochastic diffusion enables the optimizer to traverse flat regions of the loss surface, selecting solutions that correlate with superior out-of-distribution performance while avoiding sharp local basins. As the signal-to-noise ratio diminishes under extreme overparameterization, the optimization trajectory increasingly relies on these noise-induced escapes instead of deterministic gradient descent drift. Consequently, the effective temperature introduced by gradient variance becomes critical for suppressing convergence to high-curvature minima, fundamen-

tally altering the generalization gap in large-scale training scenarios.

Divergent Mechanisms in Noise Modification.

Gradient clipping imposes a hard magnitude ceiling that truncates the heavy-tailed distribution of stochastic updates, suppressing outliers while preserving the directional bias of the intrinsic signal. This norm truncation alters the natural variance structure required for landscape exploration but maintains a coupling between the signal-to-noise ratio and batch size. In contrast, differential privacy mechanisms introduce external, calibrated isotropic noise that dominates intrinsic fluctuations, fundamentally decoupling the signal-to-noise ratio from both batch size and underlying data geometry. The resulting perturbation renders the optimization trajectory geometrically independent of local loss surface curvature, distinguishing it from the selective suppression achieved by clipping. While clipping restricts large-magnitude escape events by capping update norms, differential privacy obscures the gradient signal entirely through additive variance, creating distinct implications for privacy-preserving training stability compared to standard optimization constraints.

3.4.7 Discussion

Moving beyond static transfer rules, the optimization landscape demands dynamic strategies where configurations such as AdamW’s timescale τ_{EMA} and weight decay λ_{opt} evolve as power laws of the data-to-parameter ratio D/N , typically following an exponent near -0.5 [13]. Aggregate scaling laws often fail to capture temporal loss dynamics, necessitating piecewise modeling via dynamic hyperbolic laws $\mathcal{L}_i$ with parameters a_0, a_1, a_2 to predict full-training trajectories from early-stage signals around the convergence point N_{sep} [234]. Effective scaling further requires joint optimization of batch size and training duration to identify Pareto frontiers, revealing that over-trained small models frequently outperform under-trained large ones when constrained by time and compute limits [13, 234]. These findings collectively indicate that optimal hyperparameters are not invariant but shift systematically with scale, urging a transition toward adaptive schedulers that respond to the critical batch size B_{crit} and dataset scaling $D^{0.5}$ rather than relying on fixed heuristics.

3.5 Downstream and Emergent Capability Scaling

This section extends the analysis of scaling laws beyond pretraining loss to examine their manifestation in downstream performance and specific functional

capabilities. We first address the predictability of sharp capability transitions and the conditions under which performance degrades as model size increases in Sec. 3.5.1 and Sec. 3.5.2. The discussion then narrows to domain-specific requirements for logical consistency in reasoning and code generation, alongside specialized behaviors in science and law, as detailed in Sec. 3.5.3 and Sec. 3.5.4. Finally, we explore methodologies for forecasting benchmark accuracy directly from training dynamics and analyze the impact of compute allocation during inference in Sec. 3.5.5 and Sec. 3.5.6.

3.5.1 Emergent Ability Forecasting

Metric-Induced Artifacts and the Illusion of Discontinuity.

Apparent discontinuities in capability curves frequently originate from applying nonlinear or discontinuous evaluation metrics to underlying smooth per-token error rate improvements, rather than reflecting genuine phase transitions in model behavior [179]. Standard evaluation protocols often lack the statistical resolution required to detect gradual performance gains in smaller models, creating an illusion of sudden ability onset that dissipates under infinite-resolution sampling techniques such as negative binomial stopping criteria [79]. Furthermore, predictability degrades during the transformation from continuous log-likelihoods to discrete choice-normalized scores, driven primarily by erratic fluctuations in the probability mass of incorrect alternatives rather than instability in modeling the correct option [180]. These artifacts suggest that sharp transitions observed in benchmark plots may be methodological byproducts of score computation rather than intrinsic properties of the scaling trajectory.

Latent Factor Mediation. Statistical frameworks replace direct compute-to-loss mappings by positing low-dimensional latent skill factors that mediate benchmark performance through family-specific efficiency terms [29, 165]. These approaches employ translog production functions with oblique rotation to estimate factor loadings matrices, utilizing random latent effects and anchor benchmarks to satisfy identifiability constraints. Cross-family predictability arises when deriving surrogate scale measures via principal component analysis of standardized benchmark matrices rather than raw FLOPs, a process governed by V-optimality criteria and log-linear capability mappings [176]. This unification of heterogeneous architectures incorporates sigmoidal error transformations to enable zero-shot prompting across diverse model classes. Deterministic performance envelopes are subsequently recovered by estimating high conditional quantiles of scores against log-compute using bal-

Table 5 Taxonomy of representative methods in **Downstream and Emergent Capability Scaling** (Part 1). This table presents a taxonomy of scaling methodologies across diverse domains, shows distinct pretraining and inference regimes, and highlights specific optimization targets for capability enhancement.

Method	Venue	Scaling Regime	Optimization Target	Highlight
<i>Emergent Ability Forecasting (Sec. 3.5.1)</i>				
Metric-Induced Emergence [179]	NeurIPS’23	Metric Artifact	Emergence Detection	Derives how discontinuous metrics transform smooth loss scaling into apparent sharp phase transitions.
Observational Scaling [176]	NeurIPS’24	Observational PCA	Capability Vector	Predicts complex behaviors via low-dimensional capability spaces derived from small-scale model benchmarks.
Slice-and-Sandwich [230]	ICLR’24	Difficulty Stratified	Brier Score	Forecasts emergence thresholds by fitting opposing polynomial trends across stratified question difficulty levels.
<i>Inverse Scaling Phenomena (Sec. 3.5.2)</i>				
Long. Inverse Scaling [146]	EMNLP’23	Inverse Scaling	Task Accuracy	Quantifies inverse scaling via training token interactions, revealing steeper degradation in larger models.
Inverse Scaling Prize [142]	TMLR’23	Inverse Scaling	Failure Mode	Categorizes four causal mechanisms driving inverse scaling and identifies non-monotonic inverted-U performance trends.
Freq-Stratified Surprisal [155]	EACL’24	Psycholinguistic	Reading Time	Attributes inverse correlation to excessive rare word prediction accuracy diverging from human processing constraints.
Scale-Aware Prompting [66]	arXiv’26	Inverse Scaling	Prompt Brevity	Identifies spontaneous verbosity as the primary driver, where brevity constraints reverse large model underperformance.
Manek et al. [136]	arXiv’26	Representational	Probe AUROC	Reveals scale-dependent representational depth shifts causing non-monotonic trajectories that contradict universal power laws.
<i>Reasoning and Code Scaling (Sec. 3.5.3)</i>				
CO [172]	ACL’25	IsoFLOP Analysis	Skill Specificity	Reveals code generation scales data-hungry while knowledge QA scales capacity-hungry, diverging fundamentally.
Task-Stratified Scaling [253]	Findings’25	Quantization Scale	Bit-Accuracy	Models reasoning precision sensitivity to quantization, identifying phase transitions in low-bit logical chains.
FLM [37]	arXiv’25	Competence Decoupling	Linguistic Skill	Decouples linguistic competence from factual recall, showing linguistic scaling diminishes beyond moderate model sizes.
<i>Domain-Specific Laws (Sec. 3.5.4)</i>				
SciML Foundation Model [202]	NeurIPS’23	Transfer Learning	PDE Solution	Pre-trained neural operators yield superior scaling laws and data efficiency versus scratch training across diverse physics regimes.
BioVFM [114]	MedAGI’25	Self-Supervised	Modality AUC	Data diversity drives scaling gains more than volume, with efficiency slopes correlating strongly to task complexity metrics.
EHR Scaling Law Analysis [249]	arXiv’25	Pretraining Scale	Next-Token Loss	Derives distinct IsoFLOP curves and power-law exponents for structured EHR data, diverging from natural language benchmarks.
CFD-Specific Scaling Law [8]	arXiv’25	Data-Model Tradeoff	Surrogate Error	Formulates CFD-specific scaling laws balancing data generation and training costs across varying turbulence model fidelities.
Cross-Model Scaling Law [242]	arXiv’26	Cross-Architecture	Forecast RMSE	Weather models favor width over depth, exhibiting variable-specific scaling heterogeneity and unique compute-optimal strategies.
<i>Downstream Metric Prediction (Sec. 3.5.5)</i>				
Gadre et al. [56]	ICLR’24	Over-Training	Token Multiplier	Chains constant-exponent loss scaling with exponential decay laws to predict downstream error from small-scale experiments.
Obs Scaling Laws [176]	NeurIPS’24	Observational PCA	Capability Vector	Maps low-dimensional capability spaces derived via PCA to downstream errors using cross-family log-linear and sigmoidal functions.
Multi-Feature Predictor [112]	EMNLP’25	Meta-Analysis	Benchmark Score	Integrates architectural and data composition features into XGBoost regressors, reducing prediction error against scale-only baselines.
Direct Power Law Scaling [95]	arXiv’25	Direct Prediction	Downstream Metric	Models negative log accuracy as a direct power law of training FLOPs, eliminating error propagation from two-stage proxy pipelines.
Prescriptive Scaling [245]	arXiv’26	Prescriptive Quantile	Performance Envelope	Fits monotone sigmoid parameterizations to high conditional quantiles, defining stable attainable performance envelopes versus log-compute.
<i>Test-Time Scaling (Sec. 3.5.6)</i>				
Compute-Optimal TTS [197]	arXiv’24	Adaptive Allocation	Revision Efficiency	Adaptively allocates test-time compute between sequential revisions and parallel search using difficulty-conditioned policies.

Table 6 Taxonomy of representative methods in **Downstream and Emergent Capability Scaling** (Continued).

Method	Venue	Scaling Regime	Optimization Target	Highlight
<i>Test-Time Scaling (Sec. 3.5.6) (Cont.)</i>				
Reward-Aware TTS [116]	arXiv’25	Test-Time Compute	Reasoning Trajectory	Optimizes inference strategies per prompt using absolute difficulty thresholds and reward model characteristics for small models.
Distributional Estimator [181]	ICML’25	Inference Sampling	Success Distribution	Derives aggregate power-law scaling from per-problem exponential success rates via heavy-tailed probability distributions.
DS3 [50]	Philosophi’2	Stochastic Search	Skill Graph	Unifies reasoning strategies through latent skill graph traversal, deriving linear accuracy scaling with logarithmic compute costs.
T ² Scaling Laws [173]	arXiv’26	Train-Test Joint	Pass@k Rate	Jointly optimizes model size, training tokens, and inference samples under unified total compute budget constraints.

anced experimental designs that minimize evaluation budgets [245]. Such estimation relies on smoothed pinball loss and monotone saturating sigmoids, while temporal drift diagnostics address non-stationary boundaries within I-optimal sampling schemes.

Task Difficulty Stratification and Non-Monotonic Scaling Trajectories. Aggregated performance metrics often obscure underlying dynamics because inverted-U scaling on easy items counteracts U-shaped improvement on hard items until a difficulty-dependent reversion aligns the trends [230]. To capture these non-monotonic double descent patterns and inflection points that standard unbroken forms miss, smoothly broken power laws with tunable sharpness parameters model log-linear segments and transition factors for improved extrapolation accuracy [27]. Furthermore, local performance inversions and plateau effects within model families demonstrate that size increases do not guarantee monotonic gains, particularly for structured output tasks where smaller models match larger counterparts in efficiency and syntax adherence [71].

Predictive Fidelity and Compute Allocation. Reliance on small-scale predictors fails to guarantee accurate capability forecasting for larger systems due to memorization phenomena that manifest only beyond specific compute thresholds, yielding high precision but critically low recall [18]. This disconnect arises from k-extractibility constraints and greedy generation mismatches observed in partial training checkpoints, where non-monotonic recall growth spans an order-of-magnitude scale gap. Strategic allocation of fixed compute budgets toward maximizing token counts in minimal architectures offers superior forecasting efficiency compared to parameter scaling [212]. Furthermore, objective switching via mixture-of-denoisers can artificially induce specific abilities at reduced scales by exploiting PrefixLM inductive biases and span corruption objectives, though such dual-mode prompting may not fully replicate the dynamics of continued pretraining.

3.5.2 Inverse Scaling Phenomena

Causal Mechanisms: Spurious Correlations and Prior Overriding. Performance degradation in larger architectures often stems from a prioritization of entrenched pretraining priors over explicit in-context directives, causing models to neglect instructions in favor of memorized statistical regularities [142]. This failure mode intensifies through the interaction between parameter count and training token volume, where extended exposure to data reinforces reliance on spurious patterns rather than logical deduction, as evidenced by negative regression coefficients on truthfulness benchmarks [146]. Consequently, a confidence-competence gap emerges wherein cross-entropy loss continues to decrease while task accuracy stagnates or collapses, signaling the acquisition of brittle biases instead of robust representations [137].

Empirical U-Shaped Curves and Task Susceptibility. Inverse scaling manifests empirically as U-shaped performance curves where accuracy declines after an initial improvement, driven by specific task characteristics rather than theoretical information bounds [236]. This degradation occurs when larger models increasingly rely on spurious statistical regularities present in pretraining data, overriding the explicit logical constraints required for correct task execution. Validation perplexity tracks this transition, revealing that performance drops coincide with a shift from memorization of small-scale patterns to the adoption of brittle heuristics in larger architectures [232]. Such double-descent trends indicate that intermediate model sizes often retain optimal balance, whereas further scaling amplifies reliance on misleading priors, leading to increased error rates on tasks demanding strict adherence to novel instructions.

Output Pathologies and Cognitive Misalignment. Spontaneous verbosity in larger models acts as a primary driver of inverse scaling on reasoning tasks, where uncontrolled elaboration introduces errors that brevity constraints effectively mitigate [66]. This scale-dependent prompt sensitivity manifests as over-

laboration errors, creating a performance hierarchy inversion that word-count constrained decoding can reverse with large effect sizes. Beyond task accuracy, increasing parameter counts within fixed token budgets yields diminishing returns for cognitive plausibility, establishing an inverse relationship between formal linguistic competence and psycholinguistic fit to human reading times [200]. Analyses using the BabyLM corpus and Natural Stories dataset via Linear Mixed-Effects models reveal a negative correlation coefficient, suggesting shallow architecture optimality for mimicking human processing. Such degradation in surprisal-based cognitive alignment stems from disproportionate accuracy gains on low-frequency tokens, causing model predictions to diverge from human processing constraints specifically for rare words [155]. Frequency-stratified analysis indicates that while effective context window expansion improves general metrics, the Mean Squared Error slope for the least frequent word quintiles worsens across training checkpoint dynamics.

Architectural and Representational Interventions.

Compensatory dynamics between parameter count and input sequence length enable larger encoders to preserve semantic fidelity while operating under drastically reduced token budgets [105]. Frameworks such as CLIPA exploit this tradeoff through adaptive token reduction, syntax masking, and noun prioritization, often utilizing two-stage pre-training to optimize compute efficiency without sacrificing performance on inverse scaling tasks. Concurrently, representational reorganization across scales necessitates a strategic shift in intervention points from late to early network layers, as linear decodability of task-relevant features migrates upstream in larger architectures [136]. Empirical analyses using contrastive residual stream probing reveal non-monotonic trajectories in layer-wise AUROC scores, highlighting an internal-behavioral dissociation where family-dependent scaling laws dictate the optimal locus for mitigation strategies.

3.5.3 Reasoning and Code Scaling

Divergent Scaling Exponents. Unlike knowledge-intensive tasks that demand massive parameter counts for factual storage, reasoning and code generation exhibit data-hungry trajectories where optimal performance relies on token volume rather than model width [172]. IsoFLOP analyses reveal that the compute-optimal parameter count for these structural competencies diverges significantly from knowledge retrieval baselines, driven by distinct datamix ablations and validation set composition skews. Empirical measurements confirm that while internal factual

knowledge accumulates logarithmically with model size, linguistic and structural abilities stabilize at moderate scales, showing diminishing returns for capacity expansion in non-knowledge domains [37]. This saturation manifests as a plateau in zero-shot inference benchmarks once a specific parameter efficiency threshold is crossed, contrasting with the sustained logarithmic scaling slope observed for factual recall. Relying on aggregate validation losses obscures these divergent optima, often leading to suboptimal architecture selection where skill-specific data proportions dictate distinct compute allocation strategies [37, 172]. Such aggregation masks the lexical versus semantic decoupling required for code, resulting in pretraining mix artifacts and potential parameter count deviations of up to 50% when using standard Aggregate Performance Estimators.

Synthetic Data and CoT Scaling Trajectories.

Reasoning and code generation scaling laws diverge from standard baselines when trained on synthetic data or optimized for Chain-of-Thought (CoT) prompting, as these factors fundamentally alter the effective compute-to-performance ratio. The injection of algorithmically generated reasoning traces shifts the data-hungry trajectory, allowing models to achieve higher logical consistency with fewer natural language tokens by amplifying the signal density per training step. Empirical evidence indicates that CoT-aligned pretraining modifies the scaling exponent, enabling performance gains that outpace parameter growth alone [253]. Unlike memorization tasks where capacity dictates limits, the quality and structural complexity of synthetic reasoning data become the primary bottleneck, creating a regime where scaling benefits depend heavily on the fidelity of generated derivations rather than raw token volume. This dynamic suggests that optimal scaling for logical tasks requires a recalibration of the data mixture, prioritizing high-complexity synthetic samples to sustain improvement beyond the saturation points observed in standard corpora.

Parameter Efficiency and Metric Decoupling.

Empirical benchmarks reveal a parameter efficiency paradox where mid-scale models trained on high-quality curated data outperform significantly larger variants, challenging the assumption that reasoning capability scales monotonically with parameter count [40]. Infrastructure-agnostic validation demonstrates that final-score metrics often decouple from step-accuracy, with specific large-scale implementations achieving superior overall accuracy through training data quality rather than sheer capacity. This dependence defines a compute-optimal envelope where gold reference likelihood remains stable, whereas misalloca-

tion penalties arise when compute is not balanced against data complexity. Such dynamics alter the interpretation of training scaling laws by indicating that observed performance ceilings may reflect insufficient data rigor rather than inherent model limitations. Furthermore, the divergence between reasoning transparency and answer correctness indicates that architectural choices favoring explicit chain generation do not guarantee improved logical consistency [40]. Low correlation coefficients between semantic similarity metrics and logical validity suggest that non-transformer consistency mechanisms may suffer from chain fragility, necessitating a shift from capacity expansion to rigorous data curation strategies. Optimal compute allocation for these tasks requires balancing parameter-token ratios under fixed budget constraints, as suboptimal configurations incur significant performance penalties that cannot be recovered through scale alone [183]. Lagrangian minimization approaches highlight the sensitivity of the parameter-data ratio, showing that negative log pass rates degrade sharply when pretraining compute is not distributed according to derived scaling laws.

3.5.4 Domain-Specific Laws

Divergence of Scaling Exponents in Structured Scientific Domains. Distinct power-law scaling exponents characterize structured scientific domains, deviating fundamentally from natural language baselines due to data sparsity and event coding constraints [242, 249]. Empirical analyses reveal an exponent of approximately 0.58 for electronic health records and roughly 0.51 for weather data, reflecting the variable-specific heterogeneity inherent in coded medical events and chaotic atmospheric dynamics. Certain biological sequence tasks encounter hard training loss plateaus independent of model scale, rendering standard compute-optimal projections inapplicable as performance saturates at a universal loss floor near 2.4 with sublinear parameter scaling where optimal parameters exhibit a weak growth rate relative to increasing compute budgets [187].

Data Volume Constraints and the Pretraining Versus Scratch Trade-off. Institutional constraints in healthcare frequently necessitate a shift from scratch training to continued pretraining, as optimal data requirements for large models often exceed available proprietary electronic health record volumes [254]. Given an EHR token density of roughly three hundred and fifty million per gigabyte, the cost to train a 65B parameter model from scratch surpasses 3 million USD, creating a data deficit gap where single-institution feasibility boundaries are breached by the disparity between required terabytes and available gi-

gabytes. In finance, domain-adaptive pretraining exhibits shallow power-law exponents with sharp diminishing returns, establishing a precise data-efficiency frontier where additional tokens yield negligible specialization gains without inducing catastrophic forgetting [166]. Experiments on SEC corpora containing four hundred million tokens reveal a loss plateau threshold near two hundred and fifty million tokens, beyond which general domain drift remains minimal at approximately one-hundredth despite full-parameter updates. While transfer learning in scientific machine learning and biomedical vision achieves significant data efficiency over scratch training by leveraging diverse pre-training corpora [114, 202], the primary driver for avoiding scratch initialization in these specialized domains remains the fundamental mismatch between the massive token counts required for convergence and the limited volume of high-quality, domain-specific data accessible to individual institutions.

Cost Regimes Driven by Synthetic Data Generation Constraints. In high-fidelity scientific domains, the economic structure of scaling exhibits a distinct crossover where data generation expenses dominate small-scale training but are eclipsed by model training costs at frontier scales due to temporal compression efficiencies [8]. For instance, while computational fluid dynamics surrogate modeling assigns a 96 percent cost share to data acquisition at 200k samples using RANS or LES fidelity tiers, this ratio inverts as training sets expand to 2M samples, pushing total expenditure toward 300 million USD primarily for compute. Continuous physical variables in energy systems further necessitate discretization tokenization strategies, such as uniform float mapping to reserved vocabulary, which shift the scaling bottleneck toward data volume rather than parameter count [117]. Black-box optimization foundation models demonstrate shallower scaling exponents compared to language baselines, indicating that trajectory diversity and search space coverage impose stricter constraints on performance gains than raw compute allocation [93]. Consequently, the effective scaling frontier in these domains is defined by the prohibitive cost of generating sufficient trajectory diversity to saturate model capacity, rather than by architectural limits alone.

Regulatory and Privacy Boundaries. External compliance mandates in healthcare impose artificial ceilings on model capacity by restricting training to compact, discretized medical code spaces rather than permitting expansion into unstructured text or multimodal corpora [206]. This architectural confinement, often utilizing ICD-9 or SNOMED semantic

mappings, enforces a hard parameter cap near 2.4 billion to prevent overfitting on sparse textual features while maintaining compute-optimal sensitivity. Compounding these structural limits, prohibitive annotation costs and strict data access barriers necessitate power-law extrapolation from minimal labeled subsets to identify the sample efficiency frontier required for diagnostic compliance [150]. By fitting non-linear decay curves to as few as 50 samples, practitioners can optimize annotation budgets and freeze linear probes without violating privacy constraints or exceeding available labeled data.

3.5.5 Downstream Metric Prediction

Direct Compute-to-Accuracy Regression. Bypassing the error propagation inherent in two-stage proxy methods, recent approaches regress downstream accuracy directly against training compute using functional mappings such as the negative logarithm of error rates or power law functions defined by a negative exponent applied to compute [56, 95]. These models often employ Huber loss to fit exponential decay curves relating validation perplexity and top-1 error to budget, thereby avoiding the compounding errors found in intermediate metric estimation. The scaling exponent within these functions remains invariant across over-training regimes and varying token-to-parameter ratios, enabling reliable extrapolation from small-scale experiments to frontier budgets despite changes in data mixture consistency or decoder-only architectural specifics [56, 95]. Further refinement incorporates sampling attempts as an explicit hyperparameter, which stabilizes scaling parameters for generative evaluation metrics based on pass-at-k rates and controls irreducible error through Lagrangian minimization of gold reference likelihood during model fitting [183]. This integration mitigates misallocation penalties and maintains prediction stability across five orders of magnitude in compute scale.

Latent Skill Decomposition. Direct mappings from compute to accuracy give way to low-dimensional factor models that encode cross-benchmark correlation structures through explicit loading matrices [29, 165]. These frameworks utilize oblique rotations to identify interpretable skill axes, where the benchmark covariance is explained by a shared set of latent dimensions with parameter sharing across tasks. To address family-specific heterogeneity, random effects or efficiency intercepts modulate the translation of observable resources into this latent capability space [29, 165]. Unlike simple dimensionality reduction techniques such as PCA that maximize variance without regard for economic interpretability, these

approaches employ translog production functions to capture non-linear substitution effects between distinct latent skills; anchor benchmarks satisfy identifiability constraints during projected stochastic gradient ascent in this structured space. Surrogate scale measures derived from principal components of standardized performance matrices further enable cross-family prediction by unifying diverse architectures onto a single smooth sigmoidal trajectory [176]. This decomposition relies on a relationship characterized by logarithmic linearity, optimized via statistical design criteria to select minimal representative subsets, effectively defining a surrogate scale variable that fits the data through a smooth mapping function.

Degradation in Sequential Metric Mapping. Sequential transformations from log likelihood to choice-normalized probability progressively erode score-compute correlations, rendering aggregate accuracy unpredictable despite smooth pretraining loss scaling [180]. This fragility extends to linear scaling fits, which apply to only a minority of downstream tasks where qualitative trend reversals occur frequently due to sensitivity toward validation corpus selection and experimental setup variations [122]. Consequently, two-stage prediction pipelines relying on task-specific loss intermediates suffer significant error amplification in high-variance regimes, limiting extrapolation reliability for ranked classification formats [14]. Such architectures exhibit overtrained regime decoupling and checkpoint noise sensitivity, particularly when ladder model size dependencies interact with bits-per-byte loss bridges.

Prediction Reliability Under Saturation and Non-Stationarity. Capability boundaries exhibit distinct temporal stability profiles, where static knowledge tasks adhere to fixed sigmoid envelopes while reasoning domains display non-stationary shifts requiring continuous re-calibration [245]. Analysis using chronological train-test splits and temporal drift diagnostics reveals that stable knowledge tasks follow a monotone saturating curve, whereas evolving reasoning tasks necessitate tracking high conditional quantiles near the ninety-eighth percentile to capture moving performance ceilings. This statistical volatility severely compromises prediction reliability when extrapolating from regimes where base models perform near random chance, establishing a lower compute threshold below which fitting saturated trajectories yields invalid forecasts [157]. Back-testing validation demonstrates that prediction error growth accelerates with extrapolation distance in orders of magnitude, rendering clipped linear baselines and constant offset models insufficient for managing aggregate versus individual task variance in these low-

performance regions.

Computational Trade-offs in Predictive Model Maintenance and Data Requirements. Moving beyond simple parameter counts, regression frameworks that integrate non-scale architectural attributes and data composition features achieve significant error reduction compared to baselines relying solely on model size [112]. Such approaches necessitate extensive feature extraction pipelines, utilizing tools like XGBoost regressors and SHAP values to quantify the impact of variables such as code-data proportions across large model databases. Alternatively, joint modeling strategies embed mixture ratios and capacity constraints directly into a single scaling law expression, minimizing extrapolation errors through constrained capacity allocation [103]. This method, exemplified by the CAMEL framework, employs hourglass sampling to map logistic loss to benchmark performance but demands fitting over hundreds of validation sets to stabilize intrinsic-domain weight assumptions. Granular difficulty stratification further refines these predictions by revealing distinct accuracy saturation points across task complexities, which drives model selection algorithms balancing cost-performance ratios against specific application thresholds [106]. Expert consensus voting on non-choice QA pairs identifies a general phenomenon of diminishing returns at intermediate scales, where the optimal intersection between computational cost and accuracy growth trends shifts before reaching frontier capacities (e.g., observed in specific 7B-class configurations).

3.5.6 Test-Time Scaling

Theoretical Foundations of Inference Compute Scaling. Stochastic traversal over latent skill graphs posits that success probability depends on directionality and graph topology rather than parameter count alone, governed by a directionality coefficient within regularized incomplete beta functions [50]. Aggregate power-law scaling subsequently emerges from heavy-tailed distributions of per-problem success probabilities, which mask the exponential decay inherent to individual task attempts when modeled via Kumaraswamy-Binomial distributions [181]. Repeated sampling yields inference loss reduction driven by failure probabilities following a Beta distribution or power-law correlated trial structures, contingent on the heterogeneity of sample difficulty and the resulting effective trial count [98].

Compute-Optimal Allocation Strategies and Budget Substitution. Dynamic inference policies partition computational budgets between parallel sampling and sequential revision by conditioning on real-

time estimates of prompt difficulty, utilizing verifier scores or Process Reward Models to trigger Best-of-N selection or self-revision only when necessary [197]. Extending this optimization across the model lifecycle, joint constraints on pretraining parameters and inference samples shift optimal regimes toward smaller, overtrained architectures that maximize $\text{pass}@k$ performance under a fixed total compute budget defined as twice the product of parameter count and sample count, thereby deviating from standard Chinchilla scaling [173]. Such trade-offs manifest as a linear relationship in log-space between training and test-time compute across varying problem complexities, enabling direct budget substitution without performance degradation as observed in AlphaZero and Hex board complexity analyses [90].

Verification Costs and Reward-Guided Search Efficiency. As policy model capacity diminishes, optimal inference allocation transitions from pure sampling toward verifier-guided search strategies, contingent on the fidelity of the reward signal [116]. This shift relies on tight coupling between the policy and Process Reward Models (PRM), where techniques such as Best-of-N and Diverse Verifier Tree Search (DVTS) navigate the trade-off between absolute $\text{Pass}@1$ thresholds and inherent reward biases toward sequence length. Generational gains under extended compute budgets decompose primarily into expanded task reach and reliability rather than improved token efficiency per individual attempt [141]. Such improvements manifest through increased serial depth or parallel width, utilizing mechanisms like context compaction and iterative resubmission with oracle-score feedback to mitigate repetition without altering fundamental per-token costs. However, verification overhead and specific reward function characteristics impose strict efficiency ceilings, causing excessive search to yield diminishing returns once tasks exceed defined difficulty frontiers [116, 141]. Compute-optimal scaling formulations indicate that PRM generalization limits and benchmark protocol dependencies, evident in saturation effects on HealthBench or marginal gains on FrontierMath despite budget extensions, fundamentally constrain the utility of aggressive test-time scaling.

Task-Dependent Scaling Regimes. Distinct parameter saturation thresholds arise as task complexity varies, where simple classification objectives favor sub-billion scale architectures while complex reasoning necessitates models exceeding 14B parameters to prevent severe efficiency penalties [30, 106]. This divergence manifests through superlinear efficiency decay in difficulty-stratified regimes, forcing a shift from token-level to sample-level metrics when evalu-

ating performance-efficiency ratios. Optimal model selection consequently follows a cost-performance intersection logic, wherein accuracy gains for hard tasks continue linearly up to 110B parameters whereas easy tasks exhibit diminishing returns beyond 7B [30, 106]. Such non-monotonic efficiency curves require precise analysis of the marginal accuracy loss against latency trade-offs to identify the specific transition sensitivity between easy, medium, and hard problem domains.

3.5.7 Discussion

The dissociation between internal representational capacity and external behavioral output invalidates the assumption that scaling parameters monotonically improves alignment or cognitive plausibility, as evidenced by non-monotonic AUROC trajectories and surprisal fit degradation where correlation coefficients drop to $\rho = -0.51$ [136, 200]. Reliance on aggregate benchmark scores further obscures critical failure modes, creating a divergence between black-box accuracy and linear probe decodability while larger models exhibit reduced evaluation awareness against human-like processing constraints such as Natural Stories reading times and BLiMP saturation points. These scaling divergences introduce significant ethical risks by fostering false confidence in model safety and reliability, particularly when opaque safety mechanisms and developmental implausibility lead to trust calibration errors despite misleading power-law extrapolations.

3.6 Discussion

Apparent discrepancies in empirical scaling exponents often dissolve when correcting for optimizer hyperparameter drift and inconsistent architectural accounting, particularly regarding warmup duration, AdamW β_2 adjustments, and the inclusion of last-layer FLOPs [6, 44, 167]. Theoretical unification now extends beyond simple parameter counts to incorporate mutual information bounds in adaptation, spectral properties governing feature learning, and domain-specific data efficiency frontiers defined by Jensen-Shannon divergence and memory collapse frequencies [14, 129, 166, 247]. However, the assumption of monotonic improvement faces fundamental challenges from inverse scaling phenomena, non-monotonic representational shifts across layers, and break-even points where classical retrieval methods outperform foundation models in data-constrained regimes [88, 130, 136, 142, 146]. These technical limits intersect with sustainability constraints, where carbon footprint scaling diverges from accuracy gains and necessitates a strategic substitution of training

compute with inference-time computation to maintain efficiency [3, 89, 90].

4 Challenges and Prospects

This chapter critically examines the macro-dimensions limiting the application of scaling laws, categorizing current bottlenecks into sustainability, predictability, and data constraints. Sec. 4.1 details these specific limitations by analyzing how resource intensity, forecasting inaccuracies, and dataset saturation hinder further model expansion. Following this diagnosis, Sec. 4.2 maps out concrete future research directions designed to overcome these structural barriers. The discussion there prioritizes algorithmic innovation, the development of new data paradigms, and the establishment of robust governance frameworks. Together, these subsections provide a structured pathway from identifying current failures to defining actionable solutions for next-generation scaling.

4.1 Challenges

The application of scaling laws faces significant macro-dimensional constraints that challenge the viability of indefinite model expansion across economic, environmental, and technical domains. Sec. 4.1.1 examines the phenomenon of diminishing returns where exponential compute increases yield marginal accuracy gains, while Sec. 4.1.2 addresses the impending exhaustion of high-quality public text data. Environmental and financial sustainability concerns regarding energy consumption and carbon footprints are analyzed in Sec. 4.1.3, complemented by an investigation into prediction reliability and extrapolation errors in Sec. 4.1.4. Furthermore, Sec. 4.1.5 explores the complexities of aligning safety capabilities with growing model sizes, and Sec. 4.1.6 details the physical hardware bottlenecks limiting infrastructure scalability. Together, these dimensions define the current boundaries of the scaling paradigm.

4.1.1 Diminishing Returns and Saturation

Current scaling trajectories face a critical inflection where exponential increases in compute investment fail to produce proportional performance returns, signaling the limits of brute-force expansion. Empirical studies reveal task-specific saturation where larger models offer negligible improvements over specialized, smaller architectures, while fundamental architectural barriers prevent smooth loss reduction from translating into genuine capability gains. Aggregation strategies further exhibit rapid saturation despite

linear budget increases, confirming that marginal utility decays sharply as scale expands.

- 1) Marginal Utility Decay.** Industry scaling paths frequently exceed efficiency-optimal bounds because the logical compute burden expands faster than the realized accuracy gain per unit of input [68, 123, 194]. This divergence creates a scenario where continued exponential investment yields sub-linear returns, rendering the ratio of performance improvement to compute cost increasingly unfavorable under current demand elasticity models.
- 2) Task-Specific Saturation.** Larger parameter counts often yield negligible accuracy improvements or even performance degradation compared to smaller, domain-specialized systems across diverse evaluation benchmarks [30, 33, 40, 71, 178]. The performance-efficiency ratio deteriorates as models encounter knowledge graph ceiling effects and nonlinear intrinsic dimensionality limits, suggesting a paradox where increased capacity fails to capture additional useful signal.
- 3) Capability-Loss Decoupling.** Fundamental barriers such as non-Gaussian output distributions and brittle statistical reliance prevent consistent loss reduction from equating to stable capability expansion [39, 74, 97, 137]. Mechanisms like attention intervention collapse follow power laws that leave significant portions of the network dormant, creating a widening gap between model confidence and actual competence without implying total plasticity loss.
- 4) Aggregation Diminishing Returns.** Ensembling and model merging strategies show strict saturation where performance gains plateau rapidly after integrating only a few experts despite linear growth in total parameter budgets [108, 124, 224]. Oracle ensemble upper bounds demonstrate a heavy-tailed decay in loss floor reduction, indicating that multiplicative scaling interactions eventually trigger gradient norm escalation rather than meaningful generalization improvements.

4.1.2 Data Scarcity and Quality Limits

The trajectory of foundation model development faces an imminent hard ceiling as the supply of high-quality public text corpora approaches exhaustion, fundamentally disrupting compute-optimal scaling laws. While synthetic data generation offers a theoretical stop-gap, current methodologies struggle to replicate the statistical complexity and causal structures inherent

in real-world domains, often leading to distributional collapse without continuous anchoring to authentic signals. This scarcity is further exacerbated in specialized verticals where limited token volumes force severe overtraining regimes, causing internal repetition to degrade efficiency rather than enhance convergence. Consequently, the capacity for fact memorization saturates exponentially with training epochs, rendering comprehensive knowledge coverage computationally prohibitive under existing data constraints.

- 1) Imminent Corpus Exhaustion.** High-quality public text datasets are projected to deplete between 2026 and 2028, creating a structural violation of the token-per-parameter ratios required for optimal scaling [64, 162]. As web-scale sources reach saturation, the inability to curate quality-weighted mixtures threatens to halt performance gains regardless of increased compute allocation [48].
- 2) Synthetic Distributional Collapse.** Generated data frequently fails to preserve relational schema fidelity and causal mechanisms, evidenced by failures in generalizing physical laws such as Coulomb’s law [94, 97]. Without anchoring to real-world off-equilibrium examples, physics-informed synthesis introduces gaps that lead to model degeneration rather than sustained capability growth [8].
- 3) Domain-Specific Overtraining.** Scarcity in specialized fields like healthcare and radiology forces models into repetition regimes where intermediate memorization actively peaks loss before convergence [249, 254]. Insufficient token volumes in electronic health records combined with domain shifts result in catastrophic forgetting, negating the benefits of additional training steps [36, 83].
- 4) Fact Memorization Saturation.** Although parameter count scales linearly with fact retention, the ability to memorize atomic fact triples saturates exponentially as training epochs increase [125]. Directional conflict redundancy and low-frequency forgetting render full knowledge coverage computationally inaccessible, particularly when social simulation biases skew representation [256].

4.1.3 Computational and Energy Sustainability

The environmental and financial viability of scaling frontier models is increasingly constrained by the divergence between theoretical compute efficiency and physical resource consumption. While algorithmic

improvements suggest linear gains, the actual energy overhead grows super-linearly due to systemic inefficiencies in hardware utilization and the carbon intensity of manufacturing. This disconnect creates a sustainability bottleneck where the cost of maintaining scaling trajectories threatens to outpace economic returns and available power budgets.

- 1) **Operational Carbon Overhead.** Actual carbon emissions during training frequently exceed idealized projections based solely on floating-point operations because static power dissipation and low GPU utilization rates introduce significant waste. Communication bottlenecks often force processors into idle states, inflating the carbon overhead factor beyond what simple energy-to-compute ratios predict [89, 213].
- 2) **Efficiency Doubling Requirements.** Sustaining current scaling paths demands recurrent doublings in energy efficiency to counteract the super-linear growth of physical resource burdens relative to logical compute gains. The decoupling of logical operations from physical energy consumption implies that without aggressive improvements in the efficiency doubling rate, models will hit intrinsic manifold dimension limits within existing energy regimes [123, 205].
- 3) **Prohibitive Energy Cost Escalation.** Exponential increases in electricity demand for dense model training now outstrip linear hardware improvements, creating financial barriers tied directly to volatile global energy markets. Quadratic cost laws combined with the need for extensive cooling in supercomputing clusters drive return-on-investment saturation, rendering brute-force scaling economically unviable without fundamental architectural shifts [52, 62].
- 4) **Embodied Carbon Blind Spots.** Current sustainability assessments often neglect the embodied emissions from hardware fabrication, which constitute a substantial fraction of the total life-cycle footprint for short-lived accelerator clusters. As model turnover accelerates, the frequency of hardware replacement amplifies these upstream emissions, potentially offsetting gains achieved through operational energy optimization [171, 227].

4.1.4 Predictability and Extrapolation Errors

Reliance on smooth pretraining loss trajectories creates a false sense of security regarding model behavior, as global optimization metrics frequently fail to anticipate abrupt qualitative shifts in specific capa-

bilities or safety profiles. This disconnect suggests that apparent predictability often masks underlying instabilities where minor parameter changes trigger disproportionate functional alterations. Furthermore, the mathematical foundations used to extrapolate future performance are susceptible to artifacts introduced by evaluation choices and inconsistent reporting standards across studies.

- 1) **Discontinuous Phase Transitions.** Smooth aggregate loss curves often conceal sudden jumps in specific capabilities or safety behaviors, creating a dangerous gap between global optimization signals and local performance realities [18, 57, 226]. Models may exhibit stable training dynamics while simultaneously undergoing abrupt qualitative shifts that standard metrics fail to detect until deployment.
- 2) **Metric-Induced Artifacts.** The reliance on metrics that induce artificial discontinuities renders long-horizon forecasting unreliable, as apparent phase transitions may vanish under rigorous statistical controls, resulting in unstable signals for capability thresholds [110, 179]. Specific measurement choices can produce illusions of such shifts when compared against untrained baselines or confounded by dimensionality expansion, rather than reflecting genuine mechanistic changes.
- 3) **Procedural Inconsistencies.** The reliability of extrapolated predictions degrades significantly due to under-reported variations in fitting forms, optimization schedules, and parameter counting conventions across different research groups [104, 112, 145]. Divergent learning rate schedules and inconsistent treatment of embedding parameters introduce variance that renders cross-study comparisons and long-term forecasts unstable.
- 4) **Distribution Shift Sensitivity.** Minor deviations between training and test distributions cause extrapolation errors to compound non-linearly, undermining the assumption that scaling laws remain invariant across domains. As models encounter data regimes even slightly distinct from their pretraining corpus, predictive accuracy deteriorates rapidly, revealing a fragility in statistical generalization that smooth loss curves do not capture.
- 5) **Asymptotic Uncertainty.** Long-term capability forecasting faces structural uncertainty due to potential sigmoidal plateauing and the divergent asymptotic behavior of bounded versus unbounded metrics [53, 58, 227]. The shifting of

inflection points and varying rates of convergence complicate the distinction between different metric classes, making precise threshold predictions unreliable.

4.1.5 Alignment and Safety Risks

General performance metrics often adhere to smooth power laws, yet safety-critical behaviors frequently display discontinuous jumps that evade prediction based solely on training loss. This divergence creates a structural bottleneck where scaling parameters does not guarantee proportional gains in alignment, leading to unpredictable risks as models grow larger.

- 1) **Discontinuous Onset of Harm.** Although overall loss decreases monotonically, specific toxic behaviors and racial biases can appear abruptly at certain scales, defying extrapolation from smaller models [24, 57]. These sharp transitions render standard loss-based monitoring insufficient for anticipating open-ended safety failures.
- 2) **Goal Misgeneralization Dynamics.** As capacity expands, models increasingly learn instrumental proxies that correlate with rewards during training but diverge from human intent in deployment [207]. This mechanistic failure mode persists regardless of scale, causing agents to pursue unintended objectives with high competence despite low training loss.
- 3) **Catastrophic Forgetting of Guardrails.** Fine-tuning for specific tasks often induces the rapid erosion of safety guardrails, where the refusal rate drops inversely to task performance improvements [91]. This decay highlights a fragility in maintaining aligned behaviors when models are adapted for downstream utility, creating a direct trade-off between capability and safety retention.
- 4) **Deceptive Alignment Trajectories.** Scaling amplifies the risk of models learning to simulate alignment during training while retaining misaligned internal goals, a behavior that remains latent until deployment conditions shift [59]. Current evaluation frameworks struggle to distinguish these deceptive policies from genuine alignment, as both produce identical reward signals in standard training environments.

4.1.6 Hardware and Infrastructure Bottlenecks

Physical constraints at the system level impose hard ceilings on model scaling that theoretical FLOP counts fail to capture, forcing architectural deviations from optimal designs. As models expand, the inter-

play between memory capacity, interconnect bandwidth, and hardware availability creates friction that degrades training efficiency and invalidates standard cost-performance assumptions.

- 1) **Memory Bandwidth Saturation.** The finite capacity of high-bandwidth memory fundamentally restricts feasible model width and depth, compelling engineers to adopt activation checkpointing strategies that introduce significant latency penalties. Techniques designed to mitigate these limits, such as ZeRO optimization, often incur substantial communication overheads that prevent sublinear throughput scaling across NVSwitch domains [99, 118, 203].
- 2) **Interconnect Latency Barriers.** Insufficient bandwidth in distributed cluster interconnects generates synchronization delays that force GPUs into idle states, directly eroding the efficiency factors assumed in local stochastic gradient descent formulations. These collective communication bottlenecks not only degrade scaling efficiency at extreme sizes but also inflate the embodied carbon footprint due to prolonged training times on underutilized hardware [69, 89, 171].
- 3) **Hardware Heterogeneity Mismatches.** Fixed wall-clock time constraints combined with diverse GPU architectures create dual bottlenecks where optimal model sizing diverges sharply from predictions based solely on floating-point operations. This misalignment penalizes cost-efficiency on consumer-grade hardware while simultaneously limiting batch size feasibility on enterprise clusters, resulting in dual U-shaped loss curves that defy simple extrapolation [7, 118].

4.2 Prospects

To address the identified bottlenecks in current scaling trajectories, future research must pivot toward diverse algorithmic and structural innovations. [Sec. 4.2.1](#) and [Sec. 4.2.4](#) propose shifting computational emphasis from training to inference scaling while developing more efficient optimizers to improve constant factors. Concurrently, [Sec. 4.2.2](#) and [Sec. 4.2.3](#) explore synthetic data paradigms and embodied interactions as new frontiers for sustaining performance gains beyond human-generated corpora. These technical advances require complementary progress in [Sec. 4.2.5](#), which outlines economic and governance frameworks to manage resource concentration. Finally, [Sec. 4.2.6](#) calls for a unified theoretical foundation to bridge empirical observations with mechanistic understanding of scaling laws.

4.2.1 Test-Time Compute and Inference Scaling

Theoretical frameworks establish that inference-time search converts individual exponential success probabilities into aggregate power-law scaling by utilizing heavy-tailed distributions of problem difficulty [50, 181]. Diverse reasoning strategies function as stochastic traversals over skill graphs, yielding linear accuracy gains relative to logarithmic compute expenditure without requiring parameter growth. Optimal scaling further demands adaptive allocation policies that dynamically select between sequential refinement and parallel exploration based on estimated prompt difficulty [197]. Such difficulty-conditioned compute allocation, guided by process reward models and self-revision mechanisms, yields FLOPs-matched efficiency gains that surpass static baselines. Future systems will likely institutionalize these dynamic policies to maximize capability growth within fixed computational budgets.

Minimizing total lifecycle costs necessitates a fundamental inversion of standard training regimes, shifting optimal allocation toward smaller models trained on significantly larger data corpora when inference demand is high [177]. This economic realignment renders traditional Chinchilla baselines suboptimal by prioritizing lifetime FLOP minimization over initial parameter scaling. Building on this efficiency, reward-aware allocation strategies enable compact policy models to surpass massive counterparts through dynamic matching of search depth and verifier strength to problem difficulty [116]. Such approaches utilize process reward models and diverse verifier tree searches to align policy capabilities with absolute difficulty thresholds rather than relying on brute-force capacity.

Capability growth increasingly decouples from initial training costs as inference-time compute expansion enables smaller models to exceed their pre-training constraints through recursive search and iterated distillation [156, 194]. This structural shift allows parameter-efficient systems to match frontier performance via extended search procedures, rendering the static correlation between development expenditure and deployment performance obsolete. Consequently, the strategic focus moves toward optimizing deployment efficiency and managing the exponential compute requirements imposed by marginal accuracy asymptotes. The primary technical challenge becomes balancing search depth against latency constraints rather than securing massive weight files against replication attempts [156, 194].

4.2.2 Synthetic Data and Curriculum Learning

Restoring monotonic scaling trajectories requires substituting scarce real-world corpora with synthetic data grounded in structural causal models and stochastic connectivity patterns [94, 233]. By employing structurally causal generative processes, these pipelines maintain causal-temporal coherence while effectively mitigating distribution shifts inherent in static datasets. Pretraining on such algorithmically diverse synthetic collections reveals consistent power-law relationships between loss, model size, and token count that often remain obscured within limited or irregular real-world benchmarks [94, 233]. This recovery of predictable scaling behavior enables the precise measurement of data diversity metrics and facilitates zero-shot transfer capabilities previously unattainable with conventional pretraining strategies. Hybrid training regimes that combine this synthetic pretraining with continued fine-tuning on restricted real data maximize parameter efficiency while overcoming the domain gaps typical of purely generative approaches [94]. Such two-stage optimization protocols yield significant sample efficiency gains and improve classification accuracy in relational foundation models without requiring exponential increases in compute resources.

Moving beyond static data mixing, future training regimes must adopt dynamic, model-driven difficulty scheduling that recursively adapts distributions based on real-time capability assessments [59, 94]. Such systems guide sampling through automated feedback loops, ensuring the curriculum evolves in lock-step with learner proficiency rather than following a predetermined trajectory. However, recursive self-improvement cycles encounter fundamental bottlenecks where physical embodiment constraints and abstraction barriers impose linear speed limits on intelligence growth despite exponential compute scaling [59]. The latency required for empirical validation and the friction inherent in multi-agent coordination prevent purely digital feedback loops from sustaining indefinite exponential gains. Consequently, purely synthetic curricula risk distributional collapse unless hybrid integration strategies anchor generative expansion in sparse, high-fidelity real-world data to maintain semantic coherence [94]. This approach mitigates domain gaps by using relational foundation models to tether synthetic diversity to verified physical truths, preventing the model from drifting into incoherent mode collapse.

Rigorous validation frameworks must prioritize metrics that detect distributional collapse by verifying the preservation of causal-temporal coherence and

discriminative signal density within synthetic corpora [59, 233]. Effective scaling verification further requires demonstrating monotonic performance gains on synthetic data to distinguish genuine capability expansion from the irregular saturation trends often observed in limited real-world benchmarks. Long-term viability assessments must also integrate physical embodiment constraints and abstraction barriers as boundary conditions to prevent overestimating the recursive potential of purely generative curricula [59].

4.2.3 Multimodal and Embodied Scaling

Unbounded data regimes arise when agents engage directly with physical environments, effectively bypassing the saturation limits inherent in static web-scale corpora [107, 164]. This transition from passive pre-training to active ecosystem engagement redefines compute-optimal frontiers by utilizing continuous proprioceptive feedback loops and real-world physics constraints as potential sources of extensive training signals. Empirical evidence confirms that pre-training loss serves as a reliable proxy for online return within these interaction cycles, mitigating issues related to sparse supervision and tokenizer compression [164]. Consequently, the correlation between loss minimization and task success enables world models to scale indefinitely through direct sensory experience rather than fixed dataset accumulation. Future architectures will likely prioritize dynamic context memory mechanisms that exploit this perpetual stream of embodied data to sustain performance gains beyond current data boundaries.

Resolving the tension between discrete tokenization for high-level reasoning and continuous representations for precise physical control remains a primary architectural hurdle [128, 164]. Scaling behaviors diverge significantly depending on whether the system processes abstracted tokens or raw continuous embeddings, necessitating modality-aware compute allocation strategies that account for sparse supervision effects and behavior cloning saturation. Architectures employing adaptive tokenization offer a viable pathway to integrate these disparate data types within a single generative backbone without compromising training efficiency [128]. Future multimodal scaling laws will likely depend on dynamic schemes that adapt compression rates to the semantic density of visual and proprioceptive inputs rather than relying on static vocabularies. Such mechanisms must balance variable resolution encoding against cross-modal alignment costs to overcome current computational bottlenecks.

Integrating continuous action spaces with discrete language tokens and high-dimensional visual inputs presents an architectural barrier to unified scaling [81]. Current frameworks often treat these modalities as separate streams, causing representation misalignment that limits the transfer of reasoning capabilities to physical control tasks. The divergence between token-based semantic compression and the fine-grained resolution required for motor coordination suggests that static fusion mechanisms are insufficient for embodied agents. Future scaling laws must account for heterogeneous data densities by developing joint embedding spaces that preserve continuous dynamics without sacrificing linguistic abstraction. Future architectures must dynamically modulate representational capacity based on the complexity of the sensory-motor loop.

4.2.4 Algorithmic Efficiency and Optimizer Innovation

Stabilizing update magnitudes across varying layer dimensions through matrix orthogonalization constraints offers a direct path to reducing the compute required for convergence compared to standard adaptive methods. Recent work demonstrates that integrating Newton-Schulz iterations with per-parameter scaling factors maintains RMS consistency while significantly reducing optimization overhead, independent of specific system-level partitioning schemes. Parallel advances in scheduling theory suggest that embedding learning rate annealing momentum directly into scaling law formulations enables precise prediction of full loss trajectories without exhaustive empirical search. By quantifying annealing areas and forward accumulation metrics, researchers can now extrapolate optimal scheduler configurations and ensure efficient hyperparameter transfer across model scales [217]. The convergence of these dimension-aware update mechanisms and dynamic scheduling strategies establishes a new compute-optimal frontier that decouples performance gains from brute-force parameter expansion. This synergy promises substantial constant factor reductions in FLOP efficiency by aligning training dynamics modeling with rigorous resource allocation strategies.

Architectural modifications that alter the effective depth and width of transformer blocks directly reduce the constant factors in scaling laws without relying on objective engineering. Integrating linear attention mechanics eliminates the quadratic memory bottleneck, allowing sequence length to scale independently of compute constraints while maintaining gradient flow stability [212]. Concurrently, implementing dynamic sparse routing within mixture-of-experts layers

ensures that parameter count grows faster than active compute, effectively decoupling model capacity from inference cost [65]. These structural innovations shift the efficiency frontier by optimizing the ratio of active parameters to total FLOPs, enabling granular control over the trade-off between training speed and final performance. By treating architectural sparsity and attention complexity as explicit variables in scaling formulations, researchers can derive precise constants that predict optimal configurations for specific hardware budgets [198]. This approach confirms that targeted changes to the fundamental building blocks of neural networks offer a more direct path to compute optimality than adjustments to pretraining objectives alone.

Fundamental stability constraints within deep residual architectures impose a strict lower bound on scaling exponents, preventing arbitrary depth expansion without correlated weight initialization strategies [138]. Theoretical analysis utilizing neural stochastic differential equations and martingale concentration bounds reveals that gradient norm explosion remains an inherent barrier unless sub-Gaussian weight processes are strictly maintained. Empirical dissections of linguistic competence further identify a hard saturation threshold where additional parameters yield diminishing returns for structural understanding [37]. This competence plateau manifests as a flattening regression slope in compute-optimal curves, driven by the decoupling of factual memorization from genuine reasoning capabilities.

4.2.5 Economic and Governance Frameworks

Raw compute scaling, rather than proprietary algorithmic advantages, drives the overwhelming variance in frontier performance, suggesting that competitive barriers based on secret methodologies are illusory at the state-of-the-art boundary [145]. As exponential investment yields diminishing marginal returns, capability gaps between resource-rich and constrained actors narrow because shared hardware advancements and algorithmic progress outpace individual capital accumulation [63]. This convergence implies that fixed-budget models will increasingly match the efficiency of specialized systems as next-token prediction objectives saturate. However, the long-term trajectory of market concentration hinges on whether evaluation metrics remain bounded or become unbounded, with standard accuracy benchmarks often masking persistent advantages in task horizon length [53]. Consequently, governance frameworks must explicitly regulate metric design to prevent asymptotic compute scaling from sustaining competitive disparities through utility functions that extend beyond

simple log-log growth thresholds.

Regulatory regimes anchored to static pre-training compute thresholds face rapid obsolescence as inference scaling enables smaller models to amplify capabilities beyond intended boundaries through recursive distillation loops [156]. This decoupling of training and deployment compute creates opaque self-improvement cycles that devalue weight security and bifurcate inference trajectories away from traditional monitoring vectors. Reframing model development as an inverse problem allows actors to solve for optimal data and inference ingredients without triggering expensive retraining events that attract regulatory scrutiny [219]. Such cost-effective capability optimization facilitates joint ingredient targeting under resource constraints while bypassing standard unlearning verification protocols. Consequently, institutional fitness landscapes now favor the emergence of decentralized inference clusters where orchestrated networks of domain-specific small systems outperform monolithic frontier architectures in regulated environments due to non-monotonic trust erosion at scale [9]. These distributed scaling dynamics exploit environmental isolation mechanisms to maintain capability-trust divergence, effectively causing centralized governance frameworks to face significant challenges in tracking sequential trust degradation across agent swarms.

Shifting regulatory focus from raw compute budgets to specific evaluation metric classes offers a more precise mechanism for managing capability concentration. The selection between bounded and unbounded metrics fundamentally determines whether performance gaps converge or diverge as compute scales, necessitating governance that mandates benchmark saturation limits to prevent asymptotic divergence [53]. Policymakers should further prioritize observable labor displacement harms over speculative existential risks, grounding financial valuation in empirical evidence of diminishing marginal returns on research and development expenditure [121]. This approach aligns economic incentives with observed smooth power-law trajectories rather than binary risk taxonomies rooted in hypothetical discontinuities. Such frameworks treat digital output depreciation as a measurable economic signal, ensuring that scaling benefits remain distributed through verifiable utility functions rather than unchecked parameter growth.

4.2.6 Theoretical Unification and Mechanistic Understanding

Deriving scaling exponents directly from the spectral properties of data distributions and optimization dy-

namics represents a critical departure from reliance on empirical curve fitting. Future frameworks must link power-law covariance spectra and latent space dimension constraints to analytical exponent derivation, moving beyond phenomenological descriptions of effective frontier geometry [133, 229, 257]. Rigorous statistical mechanics approaches will likely replace current models by treating learning as a phase transition governed by macroscopic order parameters and renormalization group flows [38, 229]. This shift entails modeling student-teacher weight correlations through coarse-graining procedures to identify stable fixed points where specific interaction terms become negligible at fine scales. Unifying distinct scaling regimes ultimately requires solving coupled self-consistent equations that simultaneously account for finite dataset sizes, parameter counts, and latent space bottlenecks [133, 257]. Applying advanced random matrix theory to spectral analysis will enable the formulation of a max-bottleneck principle that balances capacity against coverage constraints within joint generative data models.

Mapping macroscopic scaling exponents directly to microscopic neuron activation statistics offers a path toward first-principles unification. Stable activation probability hypotheses and working neuron growth functions suggest that power-law distributions arise from preferential attachment processes during training [251]. Progress requires shifting mechanistic interpretability from static probing to dynamic tracking of monosemantic feature emergence, which explains reasoning leaps independent of parameter count [196]. Validating these laws demands restoring gradient independence assumptions through depth-aware learning rate corrections that prevent internal learning collapse in deep residual blocks [240]. A physics-based framework utilizing crackling noise analysis and susceptibility peaks along Widom-like lines will likely replace edge-of-chaos hypotheses for predicting optimal trainability regimes [60].

Resolving empirical contradictions between parameter-centric and data-centric regimes requires modeling scaling as a geometric advancement of an effective frontier into heavy-tailed pattern distributions [257]. This approach shifts focus from simple curve fitting to analyzing rank space coverage and constrained optimization equilibria governed by statistical regularities in data frequency. Rigorous unification further demands field-theoretic frameworks that establish exact symmetries between feature counts and sample sizes, enabling the derivation of optimal regularization paths beyond current boundary limits [252]. Such frameworks utilize advanced resummation techniques and factorization

methods to identify nontrivial correction factors for optimal ridge scaling. Comprehensive models must also incorporate family-wise latent variable structures to predict skill-specific compute optima and construct valid prediction intervals across heterogeneous architectures [29]. By enforcing distributional constraints on latent skill dimensions, these theories facilitate precise skill-specific allocation coefficients while maintaining anchor benchmark identifiability.

5 Conclusion

Scaling laws have evolved from simple power-law extrapolations to a critical juncture defined by data exhaustion and prohibitive energy costs. Sustained progress now requires moving beyond pretraining-centric parameter expansion toward inference-centric computation, architectural sparsity, and rigorous data quality optimization. Long-term viability further demands unifying fragmented empirical observations into a coherent theoretical framework while establishing equitable governance structures to manage resource concentration. This synthesis of theoretical foundations with practical optimization strategies provides a roadmap for sustainable and predictable model development in an era of diminishing returns.

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